



Huntington Disease

Synonym: Huntington Chorea

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Summary

Clinical characteristics

Huntington disease (HD) is a progressive disorder of motor, cognitive, and psychiatric manifestations. HD typically manifests in adult life with mean age of onset around 40-50 years, although childhood onset and late onset do occur. Early manifestations include subtle changes in eye movements and coordination, minor involuntary movements, difficulty with mental planning, and often a depressed or irritable mood. As the disease progresses, chorea becomes more prominent, voluntary activity becomes increasingly difficult, dysarthria and dysphagia worsen, and intermittent outbursts of aggressive behaviors and social disinhibition may increase. In later stages, motor disability becomes severe and affected individuals are often totally dependent, mute, and incontinent. The median survival after onset is 15 to 18 years.

Diagnosis/testing

The diagnosis of HD can be established in a proband with characteristic clinical, neuroimaging, and family history findings and a heterozygous CAG trinucleotide repeat expansion in *HTT* identified by molecular genetic testing.

Management

Treatment of manifestations: Pharmacologic therapy for chorea includes typical and atypical neuroleptics, benzodiazepines, tetrabenazine, deutetrabenazine, or valbenazine. Hypokinesia and rigidity can be treated with pharmacologic and/or nonpharmacologic therapies. Sodium valproate or levetiracetam for myoclonic hyperkinesia. Selective serotonin uptake inhibitors and/or other drugs with serotonergic and noradrenergic effects for depression, anxiety, irritability, and obsessive-compulsive disorder. Neuroleptics for irritability and aggression. Apathy has been treated using amantadine, atomoxetine, methylphenidate, modafinil, bromocriptine, and bupropion. Lamotrigine and carbamazepine for mood stabilization. Mirtazapine for insomnia. Short course of zopiclone or melatonin for sleep disturbance. Nonpharmacologic therapy including

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physical therapy, speech-language therapy, psychotherapy, and cognitive behavioral therapy. Supportive care with attention to nursing needs, dietary intake, special equipment, and eligibility for state and federal benefits. Individuals with HD and family members may benefit from referral to a local HD support group for educational materials and psychological support.

Surveillance: Evaluate severity of chorea, rigidity, gait abnormalities, fine motor function, cognitive decline, functional capacity, mood and other behavioral changes (depression, irritability, anxiety, and apathy), sleep issues, and social work and family/support needs at each visit. Assessment of functional abilities using total function capacity of the Unified Huntington Disease Rating Scale is helpful.

Agents/circumstances to avoid: Levodopa-containing compounds may increase chorea; monoamine oxidase inhibitors should be avoided; alcohol and smoking are discouraged.

Genetic counseling

HD is inherited in an autosomal dominant manner. Offspring of an individual with a heterozygous CAG repeat expansion in *HTT* have a 50% chance of inheriting a CAG repeat expansion. Expansion and contraction of CAG repeat length can occur with maternal or paternal transmission; however, expansion occurs far more commonly in paternal transmission and contraction occurs more commonly in maternal transmission. Offspring who inherit a reduced-penetrance allele (36-39 CAG repeats) are at risk for HD but may not develop symptoms. Offspring who inherit a full-penetrance allele (40 or more CAG repeats) are at risk of developing HD with increased certainty assuming a normal life span. Once an HD-causing CAG repeat in *HTT* has been identified in an affected family member, predictive testing and prenatal/preimplantation genetic testing are possible. Predictive testing of asymptomatic at-risk adult family members usually involves pretest interviews in which the motives for requesting the test, the individual's knowledge of HD, the possible impact of positive and negative test results, and neurologic status are assessed. Preimplantation genetic testing exclusion protocols allow for testing of the embryo for couples in an at-risk family who do not wish to undergo predictive testing for the HD-causing allele themselves.

Diagnosis

Suggestive Findings

HD **should be suspected** in individuals with any of the following clinical and imaging findings and family history.

Clinical findings

- Progressive motor disability featuring chorea. Voluntary movement may also be affected.
- Mental disturbances including cognitive decline
- Changes in personality and/or psychiatric manifestations (depression)

Neuroimaging findings

- Progressive striatal atrophy of the caudate and putamen
- Enlargement of the lateral ventricles
- Progressive gray and white matter atrophy

Family history is consistent with autosomal dominant inheritance (e.g., affected males and females in multiple generations). Absence of a known family history does not preclude the diagnosis.

Establishing the Diagnosis

The diagnosis of HD is **established** in a proband with suggestive findings and a heterozygous CAG trinucleotide repeat expansion in *HTT* identified by molecular genetic testing (see Table 1).

CAG repeat sizes

- **Normal alleles.** 26 or fewer CAG repeats
- **Intermediate alleles.** 27 to 35 CAG repeats. An individual with an allele in this range is not typically at risk of developing manifestations of HD but, because of germline instability in the CAG tract, may be at risk of having a child with an allele in the HD-causing range. Risk estimates for germline CAG expansion have been established [Semaka et al 2013a, Semaka & Hayden 2014]. Although most individuals with an intermediate CAG repeat allele remain unaffected, rarely intermediate alleles are associated with subtle cognitive, behavioral, or motor features [Kenney et al 2007, Ha et al 2012, Killoran et al 2013, Cubo et al 2016, Savitt & Jankovic 2019, Jevtic & Provias 2020, Reguera Acuña et al 2021, Vater et al 2025].
- **Pathogenic HD-causing alleles.** 36 or more CAG repeats. Individuals who have an allele in this range are considered at risk of developing HD in their lifetime. HD-causing alleles are further classified as:
 - **Reduced-penetrance HD-causing alleles.** 36 to 39 CAG repeats. An individual with an allele in this range is at risk for HD but may not develop manifestations of HD. Asymptomatic individuals with CAG repeats in this range are common [Kay et al 2016, Ibañez et al 2024].
 - **Full-penetrance HD-causing alleles.** 40 or more CAG repeats. Alleles of this size are associated with development of HD with increased certainty assuming a normal life span.

Molecular genetic testing relies on targeted analysis to characterize the number of *HTT* CAG repeats.

Note: (1) Pathogenic *HTT* (CAG)_n repeat expansions can be detected by sequence-based multigene panels, exome sequencing, or whole genome sequencing. However, these approaches are not currently used routinely for diagnostic testing. Short-read amplicon-based *HTT* sequencing, as well as exome and genome-based approaches, are primarily used in research settings [Ciosi et al 2021, Ibañez et al 2024, Rajan-Babu et al 2024]. Long-read sequencing approaches have also been explored in research contexts [Handsaker et al 2025]. (2) Current fragment analysis diagnostic testing does not account for variation in the repeat region. Individuals with loss of CAA interruption within the CAG repeat region have an underestimated CAG repeat length by two CAGs (which may lead to a false negative genetic test result or an incorrect diagnostic category) [Findlay Black et al 2020, Dawson et al 2024, Findlay Black et al 2024]. Some individuals with estimated 34 to 35 CAG repeats using current diagnostic testing may have 36 or 37 CAG repeats and manifest features of HD [Bao et al 2023, Dawson et al 2024].

Table 1. Huntington Disease: Molecular Genetic Testing

Gene ¹	Method ^{2, 3}	Proportion of Pathogenic Variants Identified by Method
<i>HTT</i>	Targeted analysis for CAG trinucleotide repeat expansions	100%

1. See Table A. Genes and Databases for chromosome locus and protein.
2. See Table 10 for specific methods to characterize the number of CAG repeats in *HTT*.
3. Current clinical sequence-based multigene panels, exome sequencing, and genome sequencing can detect pathogenic repeat expansions in this gene but are not used routinely for diagnostic purposes.

Note: For comprehensive recommendations pertaining to predictive genetic testing for HD, see Genetic Counseling and MacLeod et al [2013].

Clinical Characteristics

Clinical Description

Huntington disease (HD) is a progressive disorder with motor, cognitive, and psychiatric manifestations. The stages of HD can be categorized into premanifest (presymptomatic, prodromal) and manifest (early, moderate, advanced) (see Figure 1). The Unified Huntington Disease Rating Scale (UHDRS) is used to assess clinical features in individuals with HD and at risk for HD [Huntington Study Group 1996]. The UHDRS is composed of six sections (motor, cognitive, behavioral, functional assessment, independence scale, and total functional capacity [TFC]). The cUHDRS, a composite measure of TFC, cognitive assessment (Stroop Color and Word Test [SCWT], Symbol Digit Modalities Test [SDMT]), and motor function (based on UHDRS total motor score) is the most sensitive measure of progression in HD [Schobel et al 2017].

The Huntington Disease Integrated Staging System (HD-ISS) provides a standardized framework for describing the full course of HD from birth through clinical motor diagnosis (see Figure 2) [Tabrizi et al 2022]. It delineates disease progression using genetic status, pathophysiologic changes, and biomarker evidence, capturing the biological onset of HD well before functional decline becomes apparent.

Premanifest

- **Presymptomatic.** Individuals with a pathogenic HD-causing allele are asymptomatic at birth. There are no clinical manifestations during childhood and adolescence in the vast majority of individuals.
- **Prodromal.** During the prodromal phase, subtle motor, cognitive, and/or psychiatric symptoms begin to emerge, preceding the diagnosis of manifest HD [Ross et al 2019, Considine et al 2025]. These subtle changes can occur as early as 15-20 years prior to the clinical onset of manifest HD.

Manifest. Once a motor diagnosis has been made, an individual is considered to have manifest HD. The TFC indicates disease stage: early, moderate, and advanced.

Assignment of clinical stage may have clinical management implications. For example, awareness of presymptomatic and prodromal HD may allow for preventive (rather than symptomatic) therapies.

Table 2. Huntington Disease: Classification and Clinical Stages

HD Classification		Features
Genetically confirmed	NOT genetically confirmed w/ family history of HD	
Presymptomatic	Clinically at risk	• No clinical motor signs/symptoms (motor DCL = 0 or 1) • No cognitive signs/symptoms • May have changes in brain imaging or other biomarkers (e.g., NEFL) ¹
Prodromal	Clinically prodromal	• Subtle motor signs (usually motor DCL = 2) AND/OR subtle cognitive signs/symptoms • Minor decline from premorbid level of function may be present but is undetectable on TFC. • Apathy, depression, or other behavioral changes may be present. • Usually changes in brain imaging

Table 2. continued from previous page.

HD Classification		Features
Genetically confirmed	NOT genetically confirmed w/ family history of HD	
Manifest	Clinically manifest ²	• Presence of clinical motor &/or cognitive signs/symptoms that have an impact on life • Functional changes (e.g., ↓ TFC) • Motor DCL = 3 or 4 (or motor DCL = 2 if cognitive changes significant AND evidence of progression)

Adapted with permission from Reilmann et al [2014]

DCL = diagnostic confidence level (from the Unified Huntington Disease Rating Scale [UHDRS]); HD = Huntington disease; NEFL = neurofilament light chain protein; TFC = total functional capacity (from the UHDRS)

1. Byrne et al [2017], Byrne et al [2018], Johnson et al [2018], Caron et al [2025]

2. Requires motor DCL = 4 AND cognitive changes

Clinical onset. The mean age of onset is approximately 45 years [Bates et al 2015, Ghosh & Tabrizi 2018]. About two thirds of individuals first present with neurologic manifestations; others present with psychiatric changes. In the early stages following diagnosis, manifestations include subtle changes in eye movements and coordination, minor involuntary movements, difficulty in mental planning, and a depressed or irritable mood (see Table 3). Affected individuals are usually able to perform most of their daily activities and continue working [Ross et al 2014, Bates et al 2015].

In approximately 25% of individuals with HD, the onset is delayed until after age 50 years, with some after age 70 years [Petracca et al 2022]. These individuals have chorea, gait disturbances, and dysphagia, but usually a more prolonged and benign course than those with earlier onset.

Table 3. Huntington Disease: Onset of Clinical Features

Onset	Clinical Feature
Early	• Clumsiness • Agitation • Irritability • Apathy • Anxiety • Disinhibition • Delusions • Hallucinations • Abnormal eye movements • Depression • Olfactory dysfunction
Moderate	• Dystonia • Involuntary movements • Trouble w/balance & walking • Chorea, twisting & writhing motions, jerks, staggering, swaying, disjointed gait (can seem like intoxication) • Trouble w/activities that require manual dexterity • Slow voluntary movements; difficulty initiating movement • Inability to control speed & force of movement • Slow reaction time • General weakness • Weight loss • Speech difficulties • Stubbornness

Table 3. continued from previous page.

Onset	Clinical Feature
Advanced	• Rigidity • Bradykinesia (difficulty initiating & continuing movements) • Severe chorea (less common) • Significant weight loss • Inability to walk • Inability to speak • Swallowing difficulties; danger of choking • Inability to care for oneself

Abnormalities of movement. Disturbances of both involuntary and voluntary movements occur [Ghosh & Tabrizi 2018]. Chorea, an involuntary movement disorder consisting of nonrepetitive, nonperiodic jerking of limbs, face, or trunk, is the major sign of HD. Chorea is present in most individuals and typically increases in severity during the first ten years. The choreic movements are continuously present during waking hours (absent during sleep), cannot be suppressed voluntarily, and are worsened by stress.

With advancing disease duration, chorea may decrease and other movement abnormalities such as bradykinesia, rigidity, and dystonia occur. Impairment in voluntary motor function is an early sign. Affected individuals and their families describe clumsiness in common daily activities. Motor speed, fine motor control, and gait are affected. Oculomotor disturbances occur early and worsen progressively. Difficulty in initiating ocular saccades, slow and hypometric saccades, and problems in gaze fixation occur in up to 75% of symptomatic individuals [Blekher et al 2006, Golding et al 2006]. Dysarthria occurs early and is common. Dysphagia occurs in the late stages. Hyperreflexia occurs early in 90% of individuals, while clonus and extensor plantar responses occur late and less frequently.

Abnormalities of cognition. A global and progressive decline in cognitive capabilities occurs in all individuals with HD. Cognitive changes include forgetfulness, slowness of thought processes, impaired visuospatial abilities, and impaired ability to manipulate acquired knowledge [Paulsen et al 2017]. These initial changes often involve loss of mental flexibility and impairment of executive functions such as mental planning and organization of sequential activities.

Memory deficits with greater impairment for retrieval of information occur early, but verbal cues, priming, and sufficient time may lead to partial or correct recall. Early in the disease the memory deficits in HD are usually much less severe than in Alzheimer disease.

The cognitive and behavioral manifestations in individuals with HD are more similar to frontotemporal dementia than Alzheimer disease. Attention and concentration are involved early [Peinemann et al 2005], resulting in easy distractibility. Language function is relatively preserved, but a diminished level of syntactic complexity, cortical speech abnormalities, paraphasic errors, and word-finding difficulties are common in late stages.

Neuropsychologic testing reveals impaired visuospatial abilities, particularly in late stages of the disease. Lack of awareness, especially of one's own disabilities, is common [Del Pino et al 2025].

Psychiatric disturbances. Prior to onset of HD, individuals tend to score high on measures of depression, hostility, obsessive-compulsiveness, anxiety, and psychoticism [Maltby et al 2021, McAllister et al 2021].

Depression is the most common psychiatric manifestation, affecting 40% of individuals with HD, followed by anxiety [Paoli et al 2017]. The incidence of depression in preclinical and symptomatic individuals is more than twice that in the general population [Clark et al 2023]. Recent neuroimaging studies revealed correlations between depressive symptoms and dysfunctional connectivity in the basal ganglia and prefrontal cortex, and

changes in limbic and paralimbic structures. Suicide and suicidal ideation are common in persons with HD, but the incidence rate changes with disease course and predictive testing results [van Duijn et al 2018, Grimaldi et al 2024]. The critical periods for suicide risk are just prior to receiving a diagnosis and later, when affected individuals experience a loss of independence [Eddy et al 2016].

Apathy is a common and disabling symptom that is related to disease stage and worsens with progression [Tabrizi et al 2013]. This is characterized by a general loss of interest, difficulty with initiating activities, and passive behavior.

Individuals with HD develop personality changes, affective psychosis, or psychosis in later stages [Gibson et al 2021]. Behavioral disturbances such as intermittent explosiveness, irritability, aggression, alcohol abuse, sexual dysfunction and deviations, and increased appetite are frequent. Delusions, often paranoid, are common. Hallucinations are less common.

Unlike the progressive cognitive and motor disturbances, the psychiatric changes tend not to progress with disease severity [Epping et al 2016].

Nutrition / weight loss. Individuals with HD tend to experience substantial weight loss, reduced muscle mass, and lower body mass index compared to unaffected individuals. These changes may reflect underlying metabolic and mitochondrial disturbances, and chorea may further contribute by increasing energy expenditure and exacerbating caloric deficits [Costa de Miranda et al 2019, Ogilvie et al 2021, Peball et al 2026]. This may begin during prodromal HD and lead to the development of a systemic body wasting syndrome. Individuals usually have very high calorie requirements to simply maintain their weight. Disturbed cholesterol metabolism that is critical for neuronal function and synaptic transmission and its reduced biosynthesis may contribute to the severe cognitive and synaptic defects observed in the disease [Kacher et al 2022].

Speech/swallowing. Individuals with HD may experience speech difficulties early in the disease and swallowing problems in later stages [Carlozzi et al 2021]. Speech impairment reflects a combination of dysarthria and word-finding difficulties, and in advanced disease some individuals may progress to complete anarthria. Swallowing difficulties arise from incoordination of oral and pharyngeal muscles, and can lead to choking episodes and aspiration pneumonia, a frequent cause of death in HD.

Sleep and circadian rhythms are disrupted in individuals with HD [Saade-Lemus & Videnovic 2023], possibly as a result of hypothalamic dysfunction and/or alterations in melatonin secretion [Cheong et al 2019, van Wamelen & Aziz 2021]. Insomnia and daytime somnolence may also be present, although this is more commonly due to psychiatric changes, depression, or chorea.

Neuroimaging. Imaging studies provide additional support for the clinical diagnosis of HD and are valuable biomarkers for both disease monitoring and clinical trials aimed at evaluating therapeutic efficacy in clinical trials [Johnson & Gregory 2019, van Eimeren et al 2023, Farag et al 2025a]. Regional atrophy of the caudate and putamen is among the earliest pathologic changes in HD [Scahill et al 2020, Scahill et al 2025], and serves as a sensitive biomarker, showing strong correlations with CAG repeat length and age at clinical onset [Fazio et al 2018, Hobbs et al 2024]. As HD progresses, cortical thinning in the occipital, motor, dorsomedial prefrontal, and parietal cortices also often occurs prior to the onset of motor manifestations [Kinnunen et al 2021]. Progressive gray and white matter atrophy occurs many years prior to predicted disease onset, and these volumes continue to decline throughout the course of the disease [Kinnunen et al 2021, Estevez-Fraga et al 2023]. In contrast, atrophy in regions like the hippocampus and cerebellum is less pronounced [Liu et al 2023].

Prognosis. As HD progresses, chorea becomes more prominent, voluntary activity becomes increasingly difficult, and dysarthria and dysphagia worsen. Most individuals are forced to give up their employment and depend increasingly on others for help, although they are still able to maintain a considerable degree of personal

independence. The impairment is usually considerable, sometimes with intermittent outbursts of aggressive behaviors and social disinhibition.

In late stages of HD, motor disability becomes severe, and the individual is often totally dependent, mute, and incontinent. The median survival time after onset is 15 to 18 years (range: 5 to >25 years). The average age at death is 54 to 58 years [Bates et al 2015].

Juvenile HD is rare and historically defined as HD with an onset of symptoms before age 20 years [Achenbach et al 2020]. The proportion of individuals with juvenile HD varies from 1% to 15% of individuals diagnosed with HD [Squitieri et al 2020, Achenbach & Saft 2021].

Although juvenile HD may have started during childhood, most individuals are diagnosed in young adulthood due to the atypical presentation and consequent clinical misdiagnosis [Achenbach et al 2020]. The term "pediatric HD" has been introduced to specifically refer to individuals who are currently affected and younger than age 18 years, since the term "juvenile HD" makes no distinction between individuals who are currently children and those with a disease onset in childhood but who are currently adults [Oosterloo et al 2024]. The motor, cognitive, and psychiatric disturbances observed in adult-onset HD are also observed in juvenile HD, but the clinical presentation is different [Oosterloo et al 2024].

In the first decade of life, individuals with juvenile HD present with common clinical manifestations of behavior disturbance and impairment of school performance, followed by dysarthria, seizures, rigidity, gait disturbances, bradykinesia, and dystonia [Cronin et al 2019]. Chorea is less frequent.

When onset occurs in the second decade, motor symptoms may resemble adult HD with a high frequency of depression, suicidal ideation, obsessive behavior, and perseveration [Fusilli et al 2018, Cronin et al 2019]. Other manifestations predominantly seen in juvenile HD include cerebellar ataxia, spasticity, tremor, and blinking and sniffing tics [Fusilli et al 2018]. Cognitive decline is most often detected by declining school performance. Juvenile HD is associated with more rapid disease progression and shorter disease duration (8-12 years) compared to adult-onset HD. The diagnosis is based on clinical judgement in combination with a family history of HD and molecular genetic testing. *HTT* CAG repeat length in individuals with juvenile HD is usually >55 CAG repeats and is often caused by anticipation, usually via paternal transmission [Oosterloo et al 2024].

Intermediate alleles. Most individuals with a CAG repeat length in the 27-35 range remain unaffected. Intermediate alleles have occasionally been associated with subtle cognitive, behavioral, or motor features [Savitt & Jankovic 2019, Vater et al 2025]. A small number of individuals with intermediate alleles in the upper CAG repeat range (34-35) are symptomatic due to sequence variation in the *HTT* CAG repeat region [Bao et al 2023, Dawson et al 2024].

Neuropathology. The primary neuropathologic feature of HD is the progressive bilateral atrophy of the caudate nucleus and putamen, with global cortical white and gray matter loss in later stages (see Figure 3) [Rüb et al 2016]. This loss is attributable to the selective degeneration of GABAergic medium spiny neurons in the caudate and putamen and select excitatory neurons of the cerebral cortex [Waldvogel et al 2015]. The preferential degeneration of enkephalin-containing medium spiny neurons of the indirect pathway of movement control in the basal ganglia precedes the loss of substance P-containing medium spiny neurons of the direct pathway, and loss of these neurons is thought to provide the neurobiological basis for chorea [Waldvogel et al 2015]. There is also evidence for loss of neurons in the globus pallidus, subthalamic nucleus, thalamus, hypothalamus, substantia nigra, and hippocampus [Vonsattel et al 1985, Vonsattel & DiFiglia 1998, Heinsen et al 1999, Petersén et al 2005, Guo et al 2012, Domínguez et al 2013, Singh-Bains et al 2016]. Region-specific patterns of neuron loss in the basal ganglia and cortex may underlie phenotypic variability of motor and psychiatric manifestations between affected individuals [Thu et al 2010, Hadzi et al 2012, Kim et al 2014, Waldvogel et al 2015, Mehrabi et al 2016]. Abnormalities are also observed in peripheral tissues such as skeletal muscle and testes, although it

remains unclear to what extent this may be secondary to pathology in the brain [Björkqvist et al 2008, van der Burg et al 2009, Chuang & Demontis 2021].

Intraneuronal inclusions containing huntingtin, the protein encoded by *HTT*, are also a prominent neuropathologic feature of the disease. However, the timing and pattern of huntingtin-containing inclusions in the brain do not correlate with the selective degeneration of the disease and are not believed to be primary determinants of pathology [Kuemmerle et al 1999, Michalik & Van Broeckhoven 2003, Arrasate et al 2004, Slow et al 2005, Slow et al 2006].

Genotype-Phenotype Correlations

A significant inverse correlation exists between the number of *HTT* CAG repeats and the age of onset of HD (see Figure 4) [Andrew et al 1993, Langbehn et al 2004, Langbehn et al 2010, Lee et al 2012] (see Molecular Genetics).

- Individuals with adult-onset HD usually have an *HTT* allele with CAG repeats ranging from 36 to 55.
- Individuals with juvenile-onset HD usually have an *HTT* allele with CAG repeats greater than 55.
- Intermediate alleles (ranging from 27 to 35 CAG repeats) have not been conclusively associated with HD but are prone to CAG repeat instability [Semaka et al 2013b].

For data on the age-specific likelihood of onset by CAG repeat length, see [ubc.ca](https://www.ubc.ca) (pdf).

A significant negative correlation exists between CAG repeat length and variability in age of onset; increased variability in age of onset is associated with smaller CAG repeat lengths, suggesting that non-CAG modifiers may have a greater effect at lower CAG repeat lengths [Langbehn et al 2004, Gusella & Macdonald 2009]. On average, CAG repeat length accounts for up to 70% of the variability in age of onset, with an estimated 10%-20% of the residual variability being accounted for by heritable factors [Li et al 2006, Gusella & Macdonald 2009, Lee et al 2012]. Many other genes have been shown to account for small amounts of this heritable portion of the variability [GeM-HD Consortium 2015, Lee et al 2022, GeM-HD Consortium 2025].

HTT CAG repeat length has been shown to predict age at death but not the duration of the illness [Keum et al 2016]. More rapid deterioration of motor, cognitive, and functional measures has been associated with larger *HTT* CAG repeat length in some studies [Aziz et al 2009, Chao et al 2017]. The progression of behavioral symptoms appears not to be related to repeat length [Ravina et al 2008].

Homozygotes for fully penetrant HD alleles appear to have a similar age of onset as heterozygotes but may exhibit an accelerated rate of disease progression [Squitieri et al 2011, Lee et al 2012].

Modifying factors on the same chromosome. Although most individuals with HD have canonical alleles (>95%), the *HTT* CAG repeat region is a hotspot for sequence variants [Goldberg et al 1995, Pêcheux et al 1995, Gellera et al 1996, Margolis et al 1999, Cattaneo et al 2025].

HTT CAG and CCG sequence variants within the CAG repeat region can be divided into loss-of-interruption (LOI) variants, associated with significantly earlier age of onset by up to approximately 13 years, and duplication-of-interruption (DOI) variants, associated with delayed onset [Ciosi et al 2019, GeM-HD Consortium 2019]. The LOI variants can be further subdivided into three categories based on the type of loss of interruption (see Figure 5).

The most common LOI variant (CAG-CCG LOI variant), which has a pure uninterrupted CAG and CCG repeat sequence, is significantly associated with faster progression of motor impairment and suggestive cognitive impairment [Dawson et al 2024]. In a longitudinal study assessing motor, cognitive, and functional decline, individuals with the CAG-CCG LOI variant showed nearly doubling of the rate of impairment of total motor score (TMS) in the UHDRS compared to individuals without this variant. Cognitive assessments showed similar

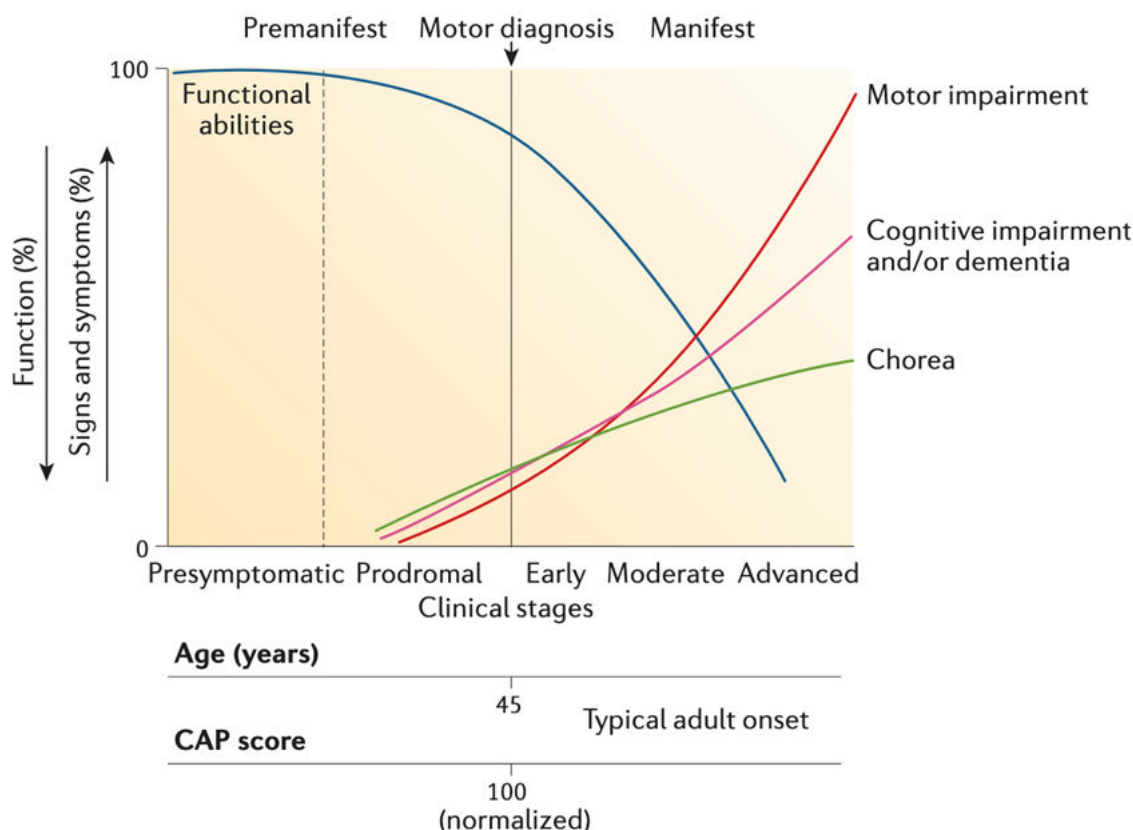


Figure 1. Natural history of Huntington disease (HD)

The premanifest period occurs before signs and symptoms of HD are identified and includes presymptomatic and prodromal phases. Presymptomatic individuals are free from signs and symptoms of HD. During the prodromal phase, subtle signs and symptoms may be present prior to the diagnosis of HD, which is usually based on motor symptoms. During manifest HD, chorea may be one of the most prominent features, followed by a slow progression of motor and cognitive impairments. Fine motor impairments (incoordination, bradykinesia, and rigidity) progress more steadily. The normalized CAG age product (CAP) score enables progression of many individuals with different *HTT* CAG repeat expansion lengths to be plotted on the same graph. Mean disease onset is at CAP score ~100 (typically age ~45 years), but there is substantial individual variability.

Reproduced from Bates et al [2015]

trends toward faster decline on the Stroop Color and Word Test (SCWT), Symbol Digit Modalities Test (SDMT), and Mini-Mental Status Exam (MMSE) [Dawson et al 2024]. Results in TRACK-HD study showed that loss of the CAA interruption was significantly associated with a faster rate of decline in total functional capacity (TFC) in the UHDRS and a trend toward faster increase in TMS [Ciosi et al 2019]. In addition to significantly earlier onset of HD and faster progression in impairment, the CAG-CCG LOI variant is also associated with accelerated atrophy of the caudate and putamen and elevated cerebrospinal fluid neurofilament light concentration (see Management, Other, **Biomarker studies**), suggesting that it promotes some of the earliest pathogenic events in the brain [Dawson et al 2024, Scahill et al 2025].

A second LOI variant, the CAG LOI variant, is defined by the loss of only the penultimate CAA codon in the CAG repeat (see Figure 5). Individuals with the CAG LOI variant have earlier onset than expected [Dawson et al 2024, GeM-HD Consortium 2025].

Modifying factors on other chromosomes. Genome-wide association studies have identified other modifier genes for HD that have since been assessed in candidate gene studies [GeM-HD Consortium 2015, Moss et al 2017]. Many of the single-nucleotide variants (SNVs) associated with variation in disease onset are in genes

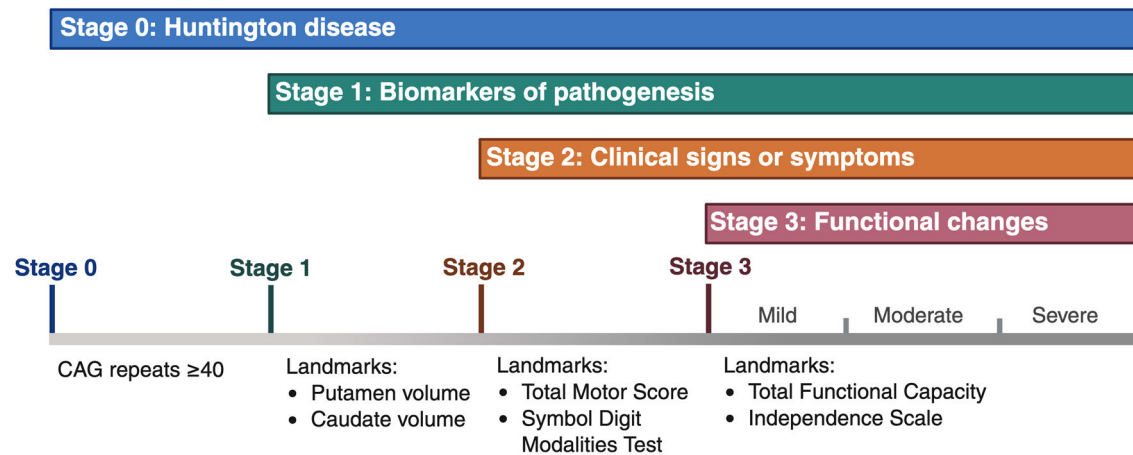


Figure 2. Representation of the Huntington Disease Integrated Staging System (HD-ISS)

This temporal representation shows the sequence of stage progression and the associated landmark assessments that define each stage entry.

Created in BioRender.

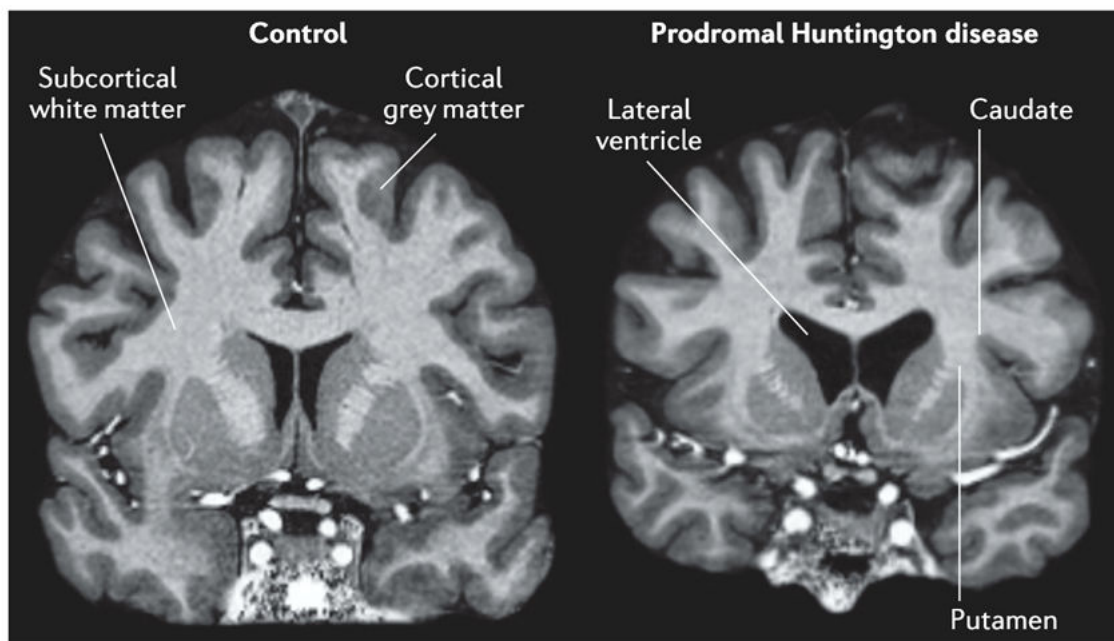


Figure 3. Brain MRI of individual with prodromal Huntington disease (HD) compared to control

The individual with prodromal HD has bilateral atrophy of the caudate and putamen and a concomitant increase in size of the fluid-filled lateral ventricle compared with control (shown using 7T MRI). There is also subtle loss of cortical gray matter volume and overall atrophy of subcortical white matter. The individual with prodromal HD had only subtle manifestations of HD that were insufficient for diagnosis of manifest HD.

Reproduced from Bates et al [2015]

involved in DNA mismatch repair (MMR). Repeatedly implicated MMR genes include *MSH3*, *MLH1*, *PMS1*, *PMS2*, and *LIG1* [GeM-HD Consortium 2015, GeM-HD Consortium 2019, Lee et al 2022, GeM-HD Consortium 2025]. In addition, the DNA repair gene *FAN1*, not previously linked to MMR, has emerged as a

strong modifier of HD onset, and its protein product has subsequently been shown to interact with the MMR pathway [Goold et al 2021, Kratz et al 2021, Porro et al 2021, Morita et al 2024].

Several of these loci have multiple independent SNVs that show significant associations with residual onset, with both delaying and hastening effects seen at the same locus [GeM-HD Consortium 2019, Lee et al 2022, GeM-HD Consortium 2025]. Many are also detected in analyses of clinical progression landmarks, including diagnostic confidence level (DCL), TFC, TMS, and SDMT [Lee et al 2022, GeM-HD Consortium 2025]. Together, these findings underscore the central role of MMR pathways in HD pathogenesis, which are thought to modify disease by altering somatic expansion through modified activity of the excision-repair complexes in a cell type-specific manner [Ciosi et al 2019, GeM-HD Consortium 2019, Iyer & Pluciennik 2021, McAllister et al 2022, Mätlik et al 2024, Pressl et al 2024].

The onset-hastening effects associated with modifiers on other chromosomes are of smaller magnitude than those associated with modifiers on the same chromosome as the CAG repeat, generally resulting in an earlier onset by approximately 0.1-5 years [GeM-HD Consortium 2015, GeM-HD Consortium 2019, Lee et al 2022, GeM-HD Consortium 2025].

Penetrance

HTT alleles with 36 to 39 CAG repeats are considered HD causing but exhibit reduced penetrance and greater variability in age of onset than alleles with repeat length in the fully penetrant range [Langbehn et al 2004]. Asymptomatic individuals with 36 to 39 CAG repeats are common [Kay et al 2016, Ibañez et al 2024].

The risk of developing HD varies between the common canonical [(CAG)*n*-CAA-CAG] interrupted repeat, observed in more than 95% of alleles, and the rare [(CAG)*n*] uninterrupted repeat, observed in about 1% of alleles [Ciosi et al 2019, GeM-HD Consortium 2019, Wright et al 2019]:

- **36 to 39 repeats.** Approximately one third of symptomatic individuals have a pure [(CAG)*n*] repeat.
- **36 and 37 repeats.** The majority of symptomatic individuals have a pure [(CAG)*n*] repeat.

Penetrance estimates for individuals with 36 to 39 CAG repeats are significantly higher in individuals from clinically ascertained families with HD than in the general population [Quarrell et al 2007, Kay et al 2016], suggesting that genetic factors, including LOI variants, modify the likelihood of manifesting HD within an average life span [GeM-HD Consortium 2015, Kay et al 2016]. An explanation for this is that loss of both CAA and CCA interruptions hastens disease onset, thereby increasing the number of individuals with CAG repeats in the reduced-penetrance range who will become symptomatic before death [Dawson et al 2024]. The CAG-CCG LOI variant is identified in ~1% of symptomatic individuals with HD but 32% of symptomatic individuals with 36 to 39 CAG repeats [Findlay Black et al 2020], showing strong enrichment at lower repeat lengths (66.7%, 45%, and 20% for CAG repeat lengths of 37, 38, and 39, respectively) and decreasing in frequency near the fully penetrant range [Dawson et al 2024].

Alleles that contain more than 40 CAG repeats are considered completely penetrant within a typical life span.

Anticipation

Anticipation occurs far more commonly in paternal transmission of an expanded *HD* allele. Anticipation arises from germline instability of the CAG repeat during spermatogenesis [Semaka et al 2013a]. Large expansions (i.e., an increase in allele size of >7 CAG repeats) occur almost exclusively through paternal transmission. Most often children with juvenile-onset HD inherit the expanded allele from their fathers, although on occasion they inherit it from their mothers [Nahhas et al 2005, Semaka et al 2015].

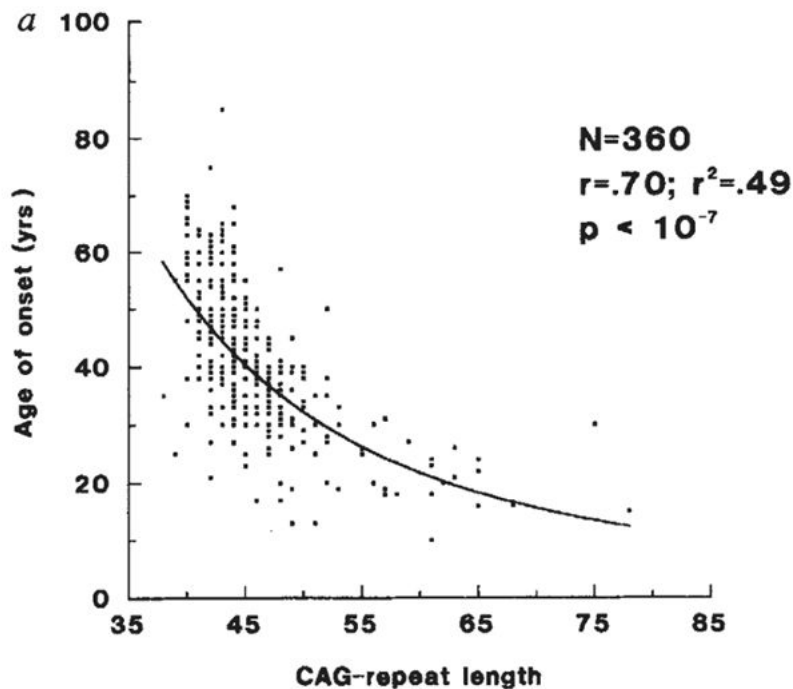


Figure 4. Age of onset of Huntington disease correlated to *HTT* CAG repeat length. The regression curve was calculated on log-transformed data.

Reproduced from Andrew et al [1993]

Nomenclature

St Vitus's dance and Sydenham's chorea. Refer to chorea in the pre-molecular genetics era

Westphal variant of HD. Refers to a rare, akinetic-rigid form of HD

Pediatric HD. Introduced to refer specifically to individuals who are currently affected and under age 18 years

Juvenile HD. Encompasses individuals who are currently affected and under age 18 years (i.e., pediatric HD) and individuals with a disease onset in childhood but who are currently adults [Oosterloo et al 2024]. Juvenile HD is historically defined as HD with an onset of symptoms before age 20 years [Achenbach et al 2020].

Huntingtin protein. The preferred term for the protein produced from an *HTT* gene that is not associated with a disease phenotype is "wild-type huntingtin (HTT)" rather than "normal" or "non-polyglutamine-expanded huntingtin." The preferred term for the protein product of an *HTT* gene that has an expanded, pathogenic CAG repeat length is "mutant huntingtin protein" (mHTT or mtHTT) rather than "variant huntingtin protein."

Prevalence

HD prevalence varies by ancestry [Kay et al 2017]. In populations of European ancestry, HD occurs at an average prevalence of 9.71:100,000 [Rawlins et al 2016], with up to 17:100,000 when multiple sources of ascertainment are accounted [Fisher & Hayden 2014, Baig et al 2016]. In contrast, HD appears less frequent in populations of East Asian ancestry, such as in Japan, China, and Korea, and in populations of African ancestry, where estimated prevalence values range from 0.1:100,000 to 2:100,000 [Pringsheim et al 2012, Xu & Wu 2015]. Further studies in Africa and East Asia with improved ascertainment are needed to confirm low reported prevalence rates in these regions. Finland has an exceptionally low prevalence of HD among European populations, possibly as a result of a population bottleneck [Sipilä et al 2015]. The prevalence of HD is increasing

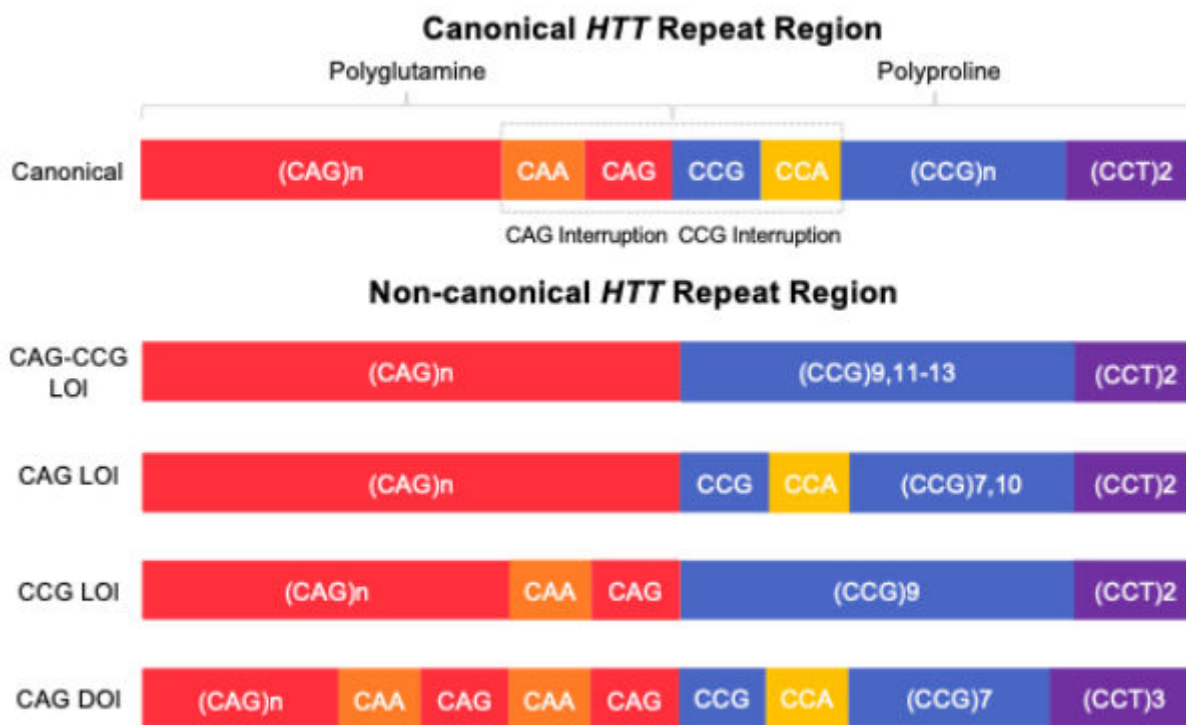


Figure 5. The *HTT* CAG and CCG sequence variants relative to the common canonical repeat region. The non-canonical sequence variants consist of the three loss-of-interruption (LOI) variants and a duplication-of-interruption (DOI) variant.

Modified from Dawson et al [2024]

in some populations in Europe, North America, and Asia, likely as a consequence of increased life expectancy and aging [Rawlins et al 2016, Lee et al 2023].

Affected individuals in the Lake Maracaibo region of Venezuela are considered to represent a founder cluster, with the highest prevalence of HD in the world [Wexler et al 2004].

The uneven distribution of HD is at least partially explained by differences in CAG repeat alleles and associated haplotypes by ancestry [Warby et al 2009, Warby et al 2011, Kay et al 2018]. The average mean CAG size in a population, as well as intermediate allele frequency, correlates with the prevalence of HD and contributes to differences in disease prevalence across major ancestry groups [Kay et al 2018].

HD alleles of 36 or more CAG repeats occur in as many as 1:400 individuals in populations of European ancestry [Kay et al 2016, Ibañez et al 2024]. The majority of these are reduced-penetrance alleles of 36-39 CAG (see Establishing the Diagnosis, **CAG repeat sizes**), suggesting that the penetrance of these alleles is low relative to larger HD alleles (0.2%-2% for 36-38 CAG alleles across a typical life span) [Kay et al 2016].

Genetically Related (Allelic) Disorders

Biallelic germline pathogenic missense variants in *HTT* are reported to cause the autosomal recessive neurodevelopmental disorder Lopes-Maciél-Rodan syndrome (LOMARS) (OMIM 617435). Individuals with LOMARS present in childhood with a Rett syndrome-like phenotype and intellectual disability [Lopes et al 2016, Rodan et al 2016, Jung et al 2021, Koshevaya et al 2022].

Differential Diagnosis

Huntington disease (HD) falls into the differential diagnosis of chorea, dementia, and psychiatric disturbances [Massey & McLauchlan 2024]. Individuals with these features and a family history suggestive of autosomal dominant inheritance who do not have an abnormal CAG repeat expansion in *HTT* are described as having an HD phenocopy. Genetic disorders characterized by phenotypes overlapping that of HD are summarized in Table 4 and reviewed elsewhere (see Schneider & Bird [2016] and Nguyen et al [2022]).

Juvenile HD. The diagnosis of HD in children is apparent in a family with a history of HD in which the father is the transmitting parent. In an affected individual with no known family history of HD and no abnormal CAG repeat length in *HTT*, ataxia-telangiectasia [Collyer & Rajan 2024], pantothenate kinase-associated neurodegeneration, Lesch-Nyhan disease [Martino et al 2013], Wilson disease [Stremmel et al 2019], progressive myoclonic epilepsy [Thakor et al 2021], and other metabolic diseases must be excluded.

Table 4. Huntington Disease: Differential Diagnosis

Gene(s)	Disorder	MOI	Features of Disorder	
			Overlapping w/HD	Distinguishing from HD
Adult-onset HD				
<i>ADCY5</i>	<i>ADCY5</i> -related movement disorder ¹	AD AR ²	• Dystonia • Chorea	• Tremor • Axial hypotonia • Speech disturbance
<i>ATN1</i>	<i>DRPLA</i> (dentatorubral-pallidoluysian atrophy)	AD	• Dystonia • Chorea • Cognitive impairment • Psychiatric manifestations	• Ataxia • Myoclonus • Epilepsy
<i>ATXN1</i> <i>ATXN2</i> <i>ATXN3</i>	Spinocerebellar ataxia type 1 (SCA1), <i>SCA2</i> , <i>SCA3</i>	AD	• Gait disturbances • Progressive motor decline • Cognitive impairment • Psychiatric manifestations	• Cerebellar ataxia • Peripheral neuropathy • Spasticity • Tremor
<i>ATXN8OS</i> / <i>ATXN8</i>	Spinocerebellar ataxia type 8 ³	AD	• Dystonia • Cognitive impairment • Chorea • Oculomotor disturbances	Ataxia
<i>C9orf72</i>	<i>C9orf72</i> frontotemporal dementia &/or amyotrophic lateral sclerosis ⁴	AD	• Dementia • Psychiatric manifestations • Dystonia	• Myoclonus • Tremor • Torticollis
<i>FTL</i>	Neuroferritinopathy (NBIA3)	AD	• Chorea • Dystonia	• Peripheral neuropathy • Ataxia • Bradykinesia • Rigidity
<i>FXN</i>	Late-onset <i>Friedreich</i> ataxia	AR	• Dysarthria • Gait disturbances	• Ataxia • Cardiomyopathy • Scoliosis • Diabetes

Table 4. continued from previous page.

Gene(s)	Disorder	MOI	Features of Disorder	
			Overlapping w/HD	Distinguishing from HD
<i>JAM2</i> <i>MYORG</i> <i>NAA60</i> <i>PDGFB</i> <i>PDGFRB</i> <i>SLC20A2</i> <i>XPRI</i>	Primary familial brain calcification	AD AR	- Chorea - Gait dysfunction - Dystonia - Speech disturbance - Neurobehavioral manifestations	- Choreoathetosis - Tremor
<i>JPH3</i>	Huntington disease-like 2	AD	Clinically indistinguishable from HD	Described exclusively in persons w/confirmed or likely African ancestry
<i>MAPT</i>	<i>MAPT</i> -related frontotemporal dementia	AD	- Late onset (mean age of onset is 49.5 yrs) - Progressive changes in motor function & memory - Behavioral & psychiatric manifestations	Not assoc w/chorea
<i>NKX2-1</i> (<i>TITF1</i>)	Benign hereditary chorea (See <i>NKX2-1-Related Disorders</i> .)	AD	Chorea	- Chorea is non-progressive - Not assoc w/dementia
<i>NOTCH2NLC</i>	<i>NOTCH2NLC</i> -related neuronal intranuclear inclusion disease ⁵	AD	- Memory loss - Personality changes - Abnormal mental behavior	- Cerebellar ataxia - Peripheral neuropathy - Seizures
<i>PRNP</i>	Huntington disease-like 1 ⁶	AD	- Progressive involuntary movements - Bradykinesia - Cognitive decline	- Early onset - Slow progression
	Gerstmann-Sträussler-Scheinker syndrome (See <i>Genetic Prion Disease</i> .)	AD	- Typically late onset (typical age at onset is early in 6th decade) - Progressive dementia - Movement disorders - Behavior changes - Psychiatric manifestations	- Faster progression - Myoclonus is a prominent involuntary movement.
<i>RFC1</i>	<i>RFC1</i> CANVAS / Spectrum Disorder	AR	- Dyskinesia - Dystonia - Chorea ⁷	- Cerebellar dysfunction - Sensory neuropathy - Vestibular areflexia
<i>RNF216</i>	<i>RNF216</i> -mediated neurodegeneration ⁸	AR	- Chorea - Behavioral manifestations - Dementia	- Hypogonadotropic hypogonadism - Cerebellar ataxia
<i>TAF1</i>	X-linked dystonia-parkinsonism (Lubag syndrome, DYT3)	XL	- Dystonia - Chorea - Bradykinesia - Rigidity - Postural instability	- Tremor - Myoclonus

Table 4. continued from previous page.

Gene(s)	Disorder	MOI	Features of Disorder	
			Overlapping w/HD	Distinguishing from HD
<i>TBP</i>	Spinocerebellar ataxia type 17	AD	• Chorea • Dementia • Psychiatric manifestations	Cerebellar ataxia
<i>VPS13A</i>	VPS13A disease (chorea-acanthocytosis)	AR	• Progressive movement disorder • Chorea • Dystonia • Cognitive impairment • Psychiatric symptoms	• Myopathy • Resting tremor • Increased serum concentration of muscle CK • Acanthocytosis • Axonal neuropathy • Seizures common • Mean age of onset ~30 yrs
<i>XK</i>	McLeod neuroacanthocytosis syndrome	XL	• Cognitive impairment • Psychiatric manifestations	• Acanthocytosis • Compensated hemolysis • McLeod blood group phenotype
Many genes	Hereditary cerebellar ataxia (See Hereditary Ataxia Overview .)	AD AR XL	Movement disorder	Prominent cerebellar & long tract signs
Juvenile HD				
<i>APTX</i> <i>SETX</i>	Ataxia w/oculomotor apraxia type 1 (AOA1) & AOA2	AR	• Gait imbalance • Dysarthria • Chorea	• Neuropathy • Oculomotor apraxia • Cerebellar ataxia
<i>ATM</i>	Ataxia-telangiectasia	AR	• Cognitive impairment • Dysphagia • Dysarthria	• Early-childhood onset • Cerebellar ataxia • Telangiectasias
<i>ATP7B</i>	Wilson disease	AR	• Personality changes • Depression • Cognitive impairment	• Elevation of liver enzymes • Tremor
<i>CP</i>	Aceruloplasminemia (NBIA10)	AR	• Chorea • Dystonia • Dysarthria • Cognitive dysfunction • Apathy	• Retinal degeneration • Diabetes mellitus
<i>FRRS1L</i>	<i>FRRS1L</i> -related Huntington disease-like chorea-dementia syndrome ⁹	AR	• Chorea • Dementia	Seizures
<i>FXN</i>	Friedreich ataxia	AR	• Dysarthria • Gait disturbances	• Ataxia • Cardiomyopathy • Scoliosis • Diabetes
<i>HPRT1</i>	Lesch-Nyhan disease ¹⁰ (See HPRT1 Disorders .)	XL	• Dystonia • Chorea	• Hyperuricemia • Self-injurious behavior (biting of fingers, hands, lips, & cheeks; head- or limb-banging)

Table 4. continued from previous page.

Gene(s)	Disorder	MOI	Features of Disorder	
			Overlapping w/HD	Distinguishing from HD
<i>HTT</i>	Lopes-Maciel-Rodan syndrome ¹¹	AR	- Dystonia - Bradykinesia - Gait disturbances	- Not assoc w/chorea - Early-childhood onset - Seizures
<i>NPC1</i> <i>NPC2</i>	Niemann-Pick disease type C ¹²	AR	- Dystonia - Chorea (less common) - Cognitive decline - Dysarthria	- Cerebellar ataxia - Vertical supranuclear gaze palsy
<i>PANK2</i>	Pantothenate kinase-associated neurodegeneration (NBIA1)	AR	- Chorea - Rigidity - Cognitive impairment - Behavioral manifestations	- Dystonia w/buccal-facial & lingual involvement - Tremor - Choreoathetosis
<i>PLA2G6</i>	<i>PLA2G6</i> -related dystonia-parkinsonism ¹³ (See <i>PLA2G6</i> -Associated Neurodegeneration [NBIA2].)	AR	- Chorea (rare) ¹⁰ - Dystonia - Rigidity - Bradykinesia - Spasticity	Resting tremor

AD = autosomal dominant; AR = autosomal recessive; CK = creatine kinase; DYT = dystonia; MOI = mode of inheritance; NBIA = neurodegeneration with brain iron accumulation; XL = X-linked

1. Menon et al [2023]

2. *ADCY5*-related dyskinesia is typically inherited in an autosomal dominant manner. Autosomal recessive inheritance has been reported in five families.

3. Lee & Bose [2025]

4. Wiesenfarth et al [2023]

5. Zhang et al [2024]

6. Huntington disease like-1 is referenced in some studies as a clinical phenotype of [genetic prion disease](#) [Moore et al 2001, Ni et al 2025]; the underlying pathologic features in reported individuals would categorize the phenotype as Gerstmann-Sträussler-Scheinker (GSS) syndrome. GSS is one of the three major phenotypes of genetic prion disease.

7. Träschütz et al [2021]

8. Santens et al [2015]

9. Madeo et al [2016]

10. Pérez-Pérez et al [2025]

11. Lopes et al [2016], Rodan et al [2016], Jung et al [2021], Koshevaya et al [2022]

12. Lee et al [2026]

13. Darling et al [2019]

Acquired conditions associated with chorea can often be excluded easily in an individual with suspected HD based on associated findings and the course of illness. Acquired causes of chorea include tardive dyskinesia, levodopa-induced dyskinesia, thyrotoxicosis, cerebrovascular disease, cerebral lupus, polycythemia, metabolic disorders, post-infective autoimmune central nervous system disorders, paraneoplastic syndromes, infectious diseases such as group A beta-hemolytic *Streptococcus*, and drugs [Nguyen et al 2022].

Management

Clinical management guidelines for Huntington disease (HD) have been published [Anderson et al 2018, Bachoud-Lévi et al 2019, Quinn et al 2020].

Evaluations Following Initial Diagnosis

To establish the extent of disease and needs in an individual diagnosed with HD, the evaluations summarized in Table 5 (if not performed as part of the evaluation that led to the diagnosis) are recommended.

Table 5. Huntington Disease: Recommended Evaluations Following Initial Diagnosis

System/Concern	Evaluation	Comment
Neurologic/ Psychiatric	- Physical exam - Neurologic assessment - Assessment of full range of motor, cognitive, & psychiatric manifestations of HD using UHDRS	The UHDRS provides reliable & consistent assessment of clinical features & progression of HD.
Genetic counseling	By genetics professionals ¹	To obtain a pedigree & inform affected persons & their families re nature, MOI, & implications of HD to facilitate medical & personal decision making
Psychosocial support & resources	- By clinicians, wider care team, & family support organizations - Note: Children & adolescents living w/a parent w/HD, sometimes in very deprived conditions, can have special challenges. Referral to local HD support group for educational material & needed psychological support is helpful (see Resources).	Assessment of family & social structure to determine need for: - Community or online resources - Social work involvement - Home nursing referral

HD = Huntington disease; MOI = mode of inheritance; UHDRS = Unified Huntington Disease Rating Scale

1. Clinical geneticist, certified genetic counselor, certified genetic nurse, genetics advanced practice provider (nurse practitioner or physician assistant)

Treatment of Manifestations

There is no cure for HD. Pharmacologic therapy is limited to symptomatic treatment [Ghosh & Tabrizi 2018, Feleus et al 2024]. Supportive care to improve quality of life, maximize function, and reduce complications is recommended (see Table 6).

Table 6. Huntington Disease: Treatment of Manifestations

Manifestation/ Concern	Treatment		Considerations/Other
Chorea	Pharmacologic	- Typical (sulpiride) & atypical (olanzapine, risperidone, quetiapine, aripiprazole) neuroleptics may partially suppress chorea - Benzodiazepines - Tetrabenazine, deutetrabenazine, ¹ & valbenazine may be assoc w/severe adverse effects. ²	Medications for chorea are prone to significant side effects incl depression, sedation, nausea, restlessness, headache, neutropenia, & tardive dyskinesia. Side effects may be worse than HD manifestations. Consider discontinuation, dose reduction, or pausing treatment for those w/ significant side effects. Persons w/mild-to-moderate chorea may be better assisted w/nonpharmacologic therapies.
	Nonpharmacologic	- PT, movement training, & OT - Speech-language therapy	

Table 6. continued from previous page.

Manifestation/ Concern	Treatment		Considerations/Other
Hypokinesia & rigidity	Pharmacologic	• Tizanidine, baclofen, & botulinum toxin injections for muscle spasms • Anti-parkinsonian agents may be helpful (specifically in juvenile HD).	• Anti-chorea medications may need to be weaned & stopped as rigidity & spasticity develop. • Anti-parkinsonian agents may also increase chorea.
	Nonpharmacologic	• PT & OT • Speech-language therapy, incl assessment of swallowing • Referral to dietician to advise on diet	
Myoclonic hyperkinesia	Sodium valproate or levetiracetam		Useful for significant isolated myoclonus
Psychiatric manifestations	Pharmacologic	• SSRIs (e.g., paroxetine, fluoxetine, citalopram, & sertraline) for depression & anxiety • Other drugs w/serotonergic & noradrenergic effects (e.g., mirtazapine & venlafaxine) for depression, anxiety, irritability, & OCD • Atypical neuroleptics² (aripiprazole, olanzapine, risperidone, & quetiapine) or typical neuroleptics² (sulpiride) for irritability & aggression. Higher doses may occasionally be needed for treatment of psychosis. • Amantadine, atomoxetine, methylphenidate, modafinil, bromocriptine, or bupropion for apathy • Lamotrigine & carbamazepine for mood stabilization • Mirtazapine for insomnia • Short-term course of zopiclone or melatonin can be useful for sleep disturbances & to address reversal of sleep-wake cycle.	• Specifically assess for suicide risk when evaluating psychiatric symptoms. • Older typical neuroleptics have greater side effects than atypical neuroleptics. • EKG prior to starting any neuroleptic • Monitor blood glucose while on neuroleptic • Using lowest possible dose is preferred to avoid side effects.
	Nonpharmacologic	Psychotherapy incl behavioral therapy for select persons w/anxiety, depression, & OCD	

Table 6. continued from previous page.

Manifestation/ Concern	Treatment	Considerations/Other
Psychosocial / social work support	• Social worker support to arrange home care when affected person can no longer care for themselves & coordinate residential or nursing home placement • HD support organizations are invaluable w/support for legal matters, employment issues, claiming entitlements, & referral to local support groups. • Assessment of nursing, diet, special equipment, & eligibility for state & federal benefits • Referral to local community mental health teams that provide psychiatric care & assessment close to home	Numerous social challenges beset persons w/HD & their families; practical help, emotional support, & counseling can provide relief. ³

OCD = obsessive-compulsive disorder; OT = occupational therapy; PT = physical therapy; SSRI = selective serotonin reuptake inhibitor

1. A deuterated analog of tetrabenazine has been modified by deuterium atom substitutions at specific sites on the molecule to increase half-life and systemic exposure [Stamler et al 2013]. These properties allow for less frequent dosing with fewer adverse effects [Frank et al 2016, Reilmann 2016].

2. Antipsychotics and vesicular monoamine transporter (VMAT) inhibitors can be associated with severe side effects including sedation, somnolence, and extrapyramidal side effects (rigidity, akathisia, and parkinsonism). Regulatory labels for VMAT inhibitors (tetrabenazine and deutetrabenazine) clearly state that it may be difficult to distinguish between progression of disease and side effects of these drugs. This must be considered when assessing clinical findings in individuals in clinical trials. When possible, lower doses of these drugs may be useful [Geva et al 2025, Tan et al 2025].

3. van Walsem et al [2024], Van Baar et al [2026]

Surveillance

To monitor existing manifestations, the individual's response to supportive care, and the emergence of new manifestations, the evaluations summarized in Table 7 are recommended [Mestre et al 2018, Considine et al 2023, Snow et al 2024].

Table 7. Huntington Disease: Recommended Surveillance

System/Concern	Evaluation	Frequency
Motor	UHDRS motor assessment ^{1, 2} & total motor score (TMS) to assess severity of chorea, rigidity, & gait abnormalities	At each visit or as indicated
	• Behavior Observation Scale Huntington (BOSH) was developed for rapid & longitudinal assessment of functional abilities in a nursing home environment.³ • Digital tools such as quantitative-motor (Q-Motor) may be helpful to avoid rater & site biases & may be particularly useful in clinical trials.⁴	Use when necessary to improve quality of care in clinical practice
Cognition	• UHDRS cognitive function,^{1, 2} verbal fluency, Symbol Digit Modalities Test (SDMT), & Stroop Color & Word Test (SCWT) • UHDRS for advanced patients (FAP)^{1, 5}	At each visit or as indicated
	• Montreal Cognitive Assessment (MoCA)⁶ to assess severity of cognitive decline; not used for screening for presence of cognitive impairment • Mini-Mental Status Exam (MMSE)⁷ to assess those w/ moderate-to-severe HD	Use when necessary to improve quality of care in clinical practice

Table 7. continued from previous page.

System/Concern	Evaluation	Frequency
Functional capacity	UHDRS - total functional capacity (TFC) score, independence scale, & functional assessment scale ¹ to describe disease progression, level of functioning, & requirements for additional caregiver aid	At each visit or as indicated
Behavioral abnormalities (depression, anxiety, aggression)	UHDRS behavioral assessment ¹ to evaluate depression, anxiety, & aggressive behavior	
	• Patient Health Questionnaire (PHQ)-9 depression measurement ⁸ • Generalized anxiety disorder (GAD)-7 anxiety measurement ⁹ • Depressive problems & anxiety problems subscales of the Youth Self-Report (YSR) ¹⁰ & Adult Self-Report (ASR) ¹¹ for at-risk persons	Use when necessary to improve quality of care in clinical practice
Stress	Responses to Stress Questionnaire – HD version (RSQ-HD) ¹²	
Sleep	• Insomnia Severity Index (ISI) ¹³ • Patient-Reported Outcomes Measurement Information System (PROMIS), Sleep Disturbance (SD); Sleep-Related Impairment (SRI) ¹⁴ • Epworth Sleepiness Scale (ESS) ¹⁵ • Pittsburgh Sleep Quality Index (PSQI) ¹⁶	
Apathy	• Dimensional Apathy Scale (DAS) ¹⁷ • Apathy Evaluation Scale (AES) ¹⁸	
Dangerous behaviors	• Columbia Suicide Severity Rating Scale Screen Version (C-SSRS-Screen) ¹⁹ • The Brøset Violence Checklist (BVC) ²⁰ • Neuropsychiatric Inventory Questionnaire (NPI-Q) ²¹	

Table 7. continued from previous page.

System/Concern	Evaluation	Frequency
Psychosocial support & resources	Assessment of need for: • Community or online resources • Social work involvement • Home nursing referral • Support needs for children living w/a parent w/HD (See Resources.)	At each visit

HD = Huntington disease; UHDRS = Unified Huntington Disease Rating Scale

1. The UHDRS is used clinically to assess HD features and functional capacity with disease progression [Huntington Study Group 1996].

2. The cUHDRS is a composite endpoint that combines the UHDRS total functional capacity (TFC), the UHDRS total motor score (TMS), the Symbol Digit Modalities Test (SDMT), and the Stroop Color and Word Test (SCWT), all of which are independently associated with HD severity [Schobel et al 2017]. This is a crucial assessment and the most sensitive measure of progression.

3. Timman et al [2005]

4. Reilmann et al [2025]

5. Youssov et al [2013]

6. Nasreddine et al [2005]

7. Folstein et al [1983]

8. Kroenke et al [2001]

9. Spitzer et al [2006]

10. Achenbach & Rescorla [2001]

11. Achenbach & Rescorla [2003]

12. Connor-Smith et al [2000]

13. Morin et al [2011]

14. Yu et al [2011]

15. Johns [1993]

16. Buysse et al [1989]

17. Radakovic & Abrahams [2014]

18. Marin et al [1991]

19. Posner et al [2011]

20. Woods & Almvik [2002]

21. Kaufer et al [2000]

Agents/Circumstances to Avoid

Levodopa-containing compounds may increase chorea.

Monoamine oxidase (MAO) inhibitors should be avoided (e.g., contraindication for tetrabenazine).

Alcohol and smoking are discouraged.

Evaluation of Relatives at Risk

See Genetic Counseling for issues related to testing of at-risk relatives for genetic counseling purposes.

Therapies Under Investigation

A wide range of potential therapeutics are under investigation in human clinical trials (see Table 8) (reviewed in Caron et al [2018], Wright et al [2020], Ferguson et al [2022], and Farag et al 2025b)). The [Huntington's Disease Research Pipeline](#) maintained by the Huntington's Disease Society of America provides a list of therapies in preclinical and clinical development. See also Figure 6 for proposed mechanisms of pharmacologic interventions.

Table 8. Huntington Disease: Pharmacologic Interventions in Clinical Trials

Intervention	Mechanism of Action
AB-1001 (BV-101)	AAV encoding for CYP46A1, enzyme converting cholesterol to 24-OH-cholesterol
ALN-HTT02	RNAi selectively targeting exon 1 <i>HTT</i> mRNA
AMT-130	rAAV5-miHTT
ATL-101	di-siRNA targeting HTT
ER2001	siRNA targeting HTT
Felodipine	Calcium channel blocker
Fenofibrate	PPAR α agonist
LPM3770164	VMAT2 inhibitor
MBF-015	Histone deacetylase 1/2 inhibitor
Metformin	Modulation of cellular energy metabolism via AMPK activation; antihyperglycemic/AMPK activator
N-acetyl cysteine (NAC)	Antioxidant
NestaCell®	Cell replacement therapy
Nicotinamide riboside	Increases NAD ⁺ levels
Nilotinib	Selective Bcr-Abl tyrosine kinase inhibitor
Pridopidine	Sigma-1 receptor activation
SKY-0515	RNA splicing molecule
SLS-005 (trehalose)	Decreases mHTT aggregation
SOM-3355 (bevantolol hydrochloride)	Beta-blocker
SRX246	Vasopressin V1a receptor blocker
Tominersen	Non-allele-selective ASO
Triheptanoin	Oral synthetic fat
VO659	CAG-targeting ASO
Votoplam (PTC518)	Small molecule splicing modulator
WVE-003	Allele-selective ASO

AAV = adeno-associated virus; AMP = adenosine monophosphate; AMPK = AMP-activated protein kinase; ASO = antisense oligonucleotide; Bcr-Abl = breakpoint cluster region–Abelson tyrosine kinase fusion protein; CYP46A1 = cytochrome P450 family 46 subfamily A member 1; di-siRNA = divalent small interfering RNA; HTT = huntingtin; mHTT = mutant huntingtin; mRNA = messenger RNA; NAD⁺ = nicotinamide adenine dinucleotide, oxidized form; PPAR α = peroxisome proliferator-activated receptor alpha; rAAV5-miHTT = recombinant AAV serotype 5 delivering microRNA targeting HTT; RNAi = RNA interference; siRNA = small interfering RNA; VMAT2 = vesicular monoamine transporter 2

ALN-HTT02 is a synthetic double-stranded RNA interference therapy designed to target exon 1 of *HTT* mRNA, leading to a nonselective reduction of huntingtin (HTT) and mutant huntingtin (mHTT) proteins. Recruitment for the clinical trial commenced in 2024, with an estimated study completion date in 2028.

AMT-130 is a microRNA targeting HTT that is delivered in an adeno-associated serotype 5 viral vector (AAV5-miHTT) directly into the striatum through a single neurosurgical procedure [Thomson et al 2023]. In a Phase I/II clinical trial, individuals that received the high dose showed no improvement after one year compared to sham control but after three years showed slowing of decline across multiple measures, including composite Unified Huntington Disease Rating Scale (cUHDRS), slowing in the decline of daily function (total functional

capacity [TFC]), and favorable biomarker changes (reduced neurofilament light chain [NEFL]). However, caution is still warranted since these results come from a small number of participants, only a portion of whom received the high dose of the drug (12 individuals), and comparisons rely on external control groups, not participants within the same trial. Although carefully matched, this kind of comparison is not as strong as a classic placebo-controlled study. Additionally, there is no report of target engagement (lowering of HTT levels), and it is unknown how this drug might impact different regions of the brain.

ATL-101 is a divalent small interfering RNA (di-siRNA) that reduces *HTT* mRNA and consequently HTT. This technology uses chemically modified oligonucleotides to leverage the RNA-induced silencing complex (RISC) and degrade a specific transcript sequence. The drug will be given by spinal injection, and early safety testing (Phase I) is expected to begin soon.

ER2001 is a genetic circuit (plasmid) encoding a neuron-targeting rabies virus glycoprotein (RVG) tag and an HTT siRNA. This circuit reprograms liver cells to transcribe and self-assemble HTT siRNA into RVG-tagged exosomes after intravenous administration. The RVG-guided HTT siRNA is further delivered through the exosome-circulating system to the cortex and striatum. A Phase I clinical trial is under way to assess safety and efficacy in individuals with early-stage HD with ascending single and multiple doses of intravenously administered ER2001.

Pridopidine is a potent and selective sigma-1 receptor (S1R) agonist. Clinical trials showed oral pridopidine is well tolerated with a benign safety profile and may have the potential to preserve functional capacity, cognition, and motor function [de Yebenes et al 2011, Reilmann et al 2019, McGarry et al 2020a, McGarry et al 2020b, Reilmann et al 2025]. In a Phase III clinical trial, sensitivity analysis in a subgroup of participants who remained off antidopaminergic medications (no VMAT2 inhibitors or antipsychotics at any time during the study) demonstrated a consistent pattern favoring pridopidine across multiple measures, including TFC and cUHDRS [Reilmann et al 2025]. Note: Antidopaminergic medications are standard treatments for psychiatric symptoms and chorea in HD. However, it is important to emphasize that their side effects, such as parkinsonism and cognitive impairment, may confound trial outcomes.

SKY-0515 is an orally administered investigational small molecule RNA splicing modifier designed to reduce concentration of HTT and PMS1, a protein previously described to be involved in driving somatic CAG repeat expansion. In a Phase I clinical trial, SKY-0515 showed reduction of mHTT at day 84 on the 9 mg daily oral dose, and additional findings include dose-dependent reductions in PMS1 mRNA, excellent brain penetration, and a favorable safety profile.

Tominersen is a non-allele-selective antisense oligonucleotide (ASO) designed to reduce concentrations of *HTT* mRNA [Tabrizi et al 2019b, McColgan et al 2023]. In a Phase I/II clinical trial, treatment with tominersen successfully targeted and reduced levels of mHTT and overall dose-dependent reductions were observed, with the greatest changes observed in the 90 and 120 mg treatment groups. The Phase III clinical trial was [discontinued in early 2021](#) due to the negative benefit/risk profile for study participants. However, analysis from the original study showed a slight benefit for a subgroup of younger individuals with less disease burden. A Phase II clinical trial focusing on 100 mg dose of tominersen is expected to conclude in 2026. The trial evaluates safety, cerebrospinal fluid (CSF) biomarkers, brain MRI changes, and function in individuals with HD.

Votoplam (PTC518) is a small molecule splicing modifier that acts via a unique mechanism to promote the inclusion of a novel pseudoexon containing a premature termination codon. This inclusion triggers nonsense-mediated decay of *HTT* mRNA, leading to a nonselective reduction of HTT [Gao et al 2024]. In a Phase II clinical trial, PTC518 lowered HTT concentration at week 12 and showed favorable safety and tolerability. In addition, at 24 months, there is a favorable dose-dependent trend on the cUHDRS, TFC, and Symbol Digit Modalities Test (SDMT) subscales in individuals with early but not later disease relative to natural history as well as dose-dependent lowering of NEFL protein concentration.

VO659 is an ASO administered intrathecally targeting mRNA produced from CAG repeats. Its HTT protein-lowering mechanism acts through steric blocking of protein translation, leading to decreased concentrations of polyglutamine proteins without degrading the mRNA transcript [Datson et al 2017]. First clinical trial data demonstrated it is generally safe and well tolerated, showing 28% mean reduction in CSF mHTT at day 85 in individuals with HD in the 40 mg dose cohort with an effect after the first dose.

WVE-003 is an allele-selective ASO that targets a single-nucleotide polymorphism (SNP3) that is commonly associated with the CAG expanded allele [Iwamoto et al 2024]. A Phase Ib/IIa clinical trial showed statistically significant, potent, and durable allele-selective silencing with 46% mean reduction in CSF mHTT compared to placebo, preservation of HTT, and generally safe and well-tolerated profile achieved in the 30 mg multidose cohort. A statistically significant correlation between mHTT lowering and slowing of caudate atrophy is an important imaging biomarker predictive of clinical outcomes.

Numerous other human clinical trials are planned or under way for HD and are listed at [huntingtonstudygroup.org](https://www.huntingtonstudygroup.org). A number of drug trials have been completed and/or are ongoing. Search [ClinicalTrials.gov](https://clinicaltrials.gov) in the US and [EU Clinical Trials Register](https://european-clinical-trials-register.eu) in Europe for information on clinical studies for a wide range of diseases and conditions.

Other

Biomarker studies. Large, multicenter observational studies TRACK-HD [Tabrizi et al 2012], PREDICT-HD [Paulsen et al 2014], and ENROLL-HD [Sathe et al 2021] have identified early clinical, imaging, and physiologic manifestations in individuals with HD. The TRACKON-HD study extended TRACK-HD to investigate compensatory mechanisms in individuals with premanifest HD [Klöppel et al 2015].

The HD Young Adult Studies (HD-YAS and HD-YAS 2.0) have provided longitudinal assessments of individuals with an *HTT* CAG repeat expansion prior to onset of HD, showing that clinical, cognitive, and neuropsychiatric measures remain stable over 4.5 years [Scahill et al 2020, Scahill et al 2025]. Despite this stability, early biological changes were evident, including elevated CSF NEFL protein concentration, reduced proenkephalin concentration, striatal atrophy, and increased somatic expansion of the *HTT* CAG repeat in blood cells [Scahill et al 2025].

Molecular biomarkers may be useful to monitor pathophysiologic changes in the central nervous system associated with HD.

NEFL, a neuron-specific intermediate filament protein, is a validated biomarker of neurodegeneration. In individuals with HD, NEFL CSF and blood concentrations increase with disease progression and correlate with clinical measures of motor and cognitive function, as well as imaging measures of brain atrophy [Byrne et al 2017, Byrne et al 2018, Johnson et al 2018, Caron et al 2025].

Glial fibrillary acidic protein (GFAP), a marker of astroglial activation, is elevated in individuals with HD and correlates with disease severity, and plasma GFAP is strongly associated with plasma NEFL across disease stages [You et al 2021, Korpela et al 2024].

Proenkephalin and prodynorphin are decreased in CSF in individuals with HD [Al Shweiki et al 2021, Niemela et al 2021, Caron et al 2022b] and provide additional insight into the health of striatal medium spiny neurons.

Mutant huntingtin (mHTT) in CSF correlates with disease stage [Byrne et al 2018] and brain mHTT levels [Southwell et al 2015, Caron et al 2021, Caron et al 2022a]; assays to measure CSF mHTT have been clinically validated [Fodale et al 2017]. CSF mHTT is currently being used as a pharmacodynamic biomarker of target engagement in HTT protein-lowering clinical trials [Niccolini et al 2015, Wagner et al 2016, Fazio et al 2020, Gutierrez et al 2020, Dong & Cong 2021, Sampedro et al 2022, Soltani Khaboushan et al 2023, Herrero-Lorenzo et al 2024] (see also Brás et al [2025]; pre-peer review, [bioRxiv](https://www.biorxiv.org)).

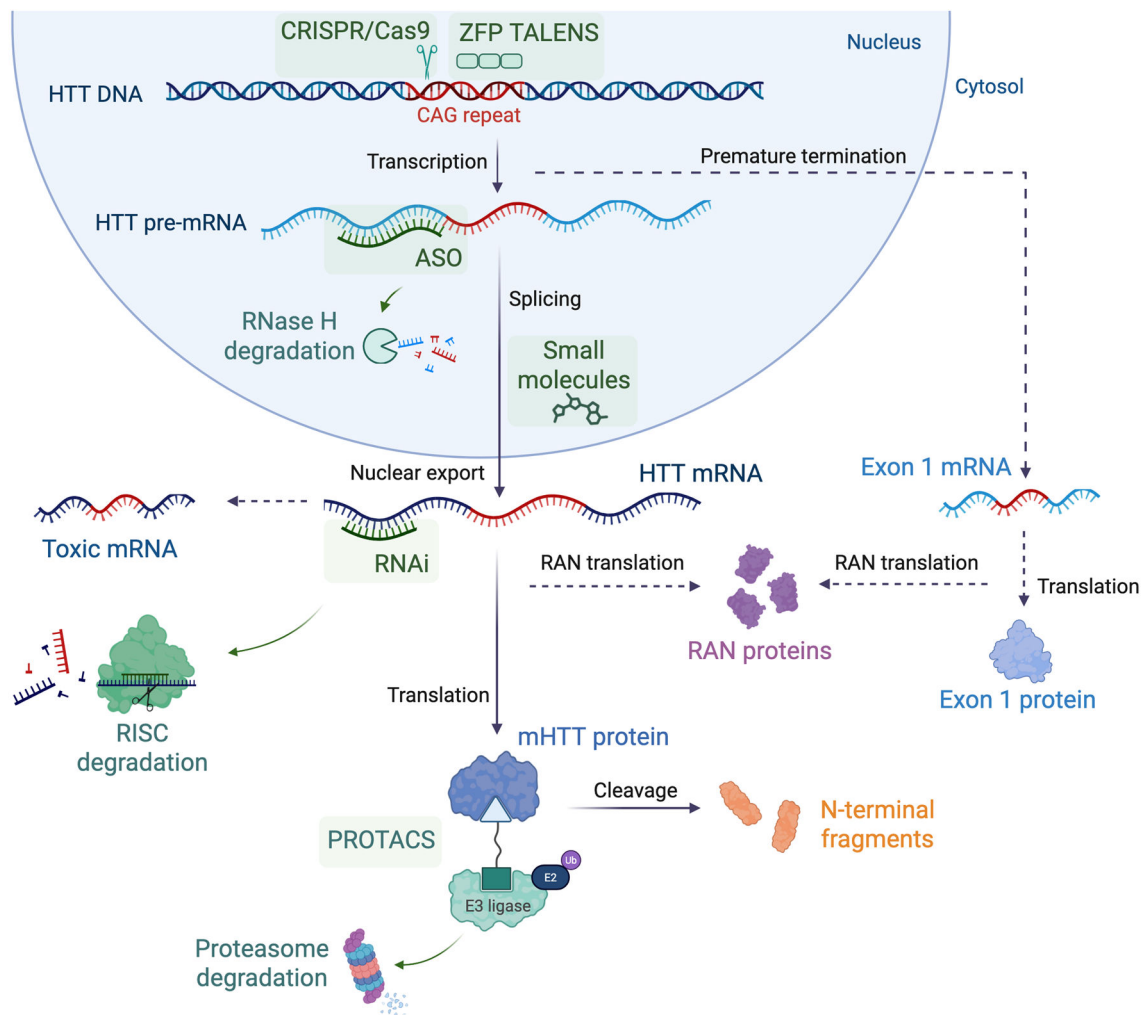


Figure 6. Mechanisms for huntingtin (HTT) lowering and production of potential toxic HTT protein species in Huntington disease (HD)

Pathogenic expanded CAG repeat tract and its polyglutamine product is in red. Therapeutic approaches are in green. Proposed mechanisms that produce alternative toxic species are represented with dotted arrows. These include mutant huntingtin protein (mHTT), exon 1 protein, N-terminal fragments, and RAN proteins.

ASO = antisense oligonucleotide; PROTACS = proteolysis-targeting chimera; RAN = repeat-associated non-ATG; RISC = RNA-induced silencing complex; TALEN = transcription activator-like effector nuclease; ZFP = zinc-finger protein

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The Huntington Disease Integrated Staging System (HD-ISS) is a clinical research tool created in 2022 to include neuroimaging with motor and cognitive scales, allowing separation of individuals with a pathogenic *HTT* CAG repeat expansion with no detectable signs of disease from those with caudate atrophy and mild cognitive impairment that do not yet meet the criteria for disease onset (see Figure 2) [Tabrizi et al 2022]. This staging system characterizes individuals in four stages: stage 0 is from birth and includes individuals with a pathogenic *HTT* CAG repeat expansion who do not have any pathologic changes; stage 1 includes individuals who have measurable indicators of underlying pathophysiology, as measured by caudate and putamen volume; stage 2 includes individuals who have a detectable clinical phenotype; stage 3 includes those individuals who demonstrate a decline in functional abilities, measured by changes in the total functional capacity and independence scales of the UHDRS. The HD-ISS staging system is particularly useful in clinical research studies including clinical trials.

Genetic Counseling

Genetic counseling is the process of providing individuals and families with information on the nature, mode(s) of inheritance, and implications of genetic disorders to help them make informed medical and personal decisions. The following section deals with genetic risk assessment and the use of family history and genetic testing to clarify genetic status for family members; it is not meant to address all personal, cultural, or ethical issues that may arise or to substitute for consultation with a genetics professional. —ED.

Mode of Inheritance

Huntington disease (HD) is inherited in an autosomal dominant manner.

Risk to Family Members

Parents of a proband

- Most individuals diagnosed with HD have an affected parent.
- An individual with HD may appear to be the only affected family member for one of the following reasons:
 - Failure to recognize the disorder in family members
 - Early death of a parent with a heterozygous *HTT* CAG trinucleotide repeat expansion before the onset of symptoms
 - The presence of an *HTT* allele with an intermediate (27-35 CAG repeats) or reduced penetrance (36-39 CAG repeats) repeat length in an asymptomatic parent (See **Offspring of a proband.**)
 - Late onset of the disease in the transmitting parent
 - Misattributed parentage
- If the proband appears to be the only affected family member (i.e., a simplex case), molecular genetic testing of the parents of the proband can be considered to determine their genetic status and inform recurrence risk assessment.

Sibs of a proband. The risk to the sibs of a proband depends on the genetic status of the proband's parents:

- If a parent has a full-penetrance HD-causing allele (40 or more CAG repeats), the risk to the sibs of inheriting a full-penetrance HD-causing allele is 50%.
- If a parent has a reduced penetrance HD-causing allele (36-39 CAG repeats), the risk to the sibs of inheriting a pathogenic HD-causing allele (of either reduced or full penetrance) is 50% (see Penetrance).
- If a parent has an intermediate *HTT* allele (27-35 CAG repeats), the risk to the sibs of inheriting a CAG repeat expansion greater than 35 repeats or a "new HD-causing allele" depends on a variety of factors, including the following (see also Anticipation):
 - **Sex of the transmitting parent.** Paternally inherited intermediate alleles are more prone to CAG expansion than maternally inherited intermediate alleles; maternal expansions of intermediate alleles into the HD range are extremely rare [Wheeler et al 2007, Semaka et al 2015].
 - **CAG repeat length.** Larger CAG repeats are more prone to expansion.
- CAG repeat length-specific estimates for repeat instability in sperm have been reported to enable genetic counselors to provide more accurate risk assessment for persons who receive an intermediate allele predictive test result [Hendricks et al 2009, Semaka et al 2013a, Semaka & Hayden 2014]. While all intermediate CAG repeat sizes were shown to have the possibility of expansion in the male germline, the probability of expansion into the pathogenic HD-causing range increases dramatically with increasing CAG repeat length. Evidence-based genetic counseling implications for intermediate allele predictive test results have been published [Semaka & Hayden 2014].

Offspring of a proband

- At conception, each child of an individual with a heterozygous CAG [repeat](#) expansion in *HTT* has a 50% chance of inheriting a CAG [repeat](#) expansion. Note: Expansion and contraction of CAG repeat size can occur with maternal or paternal transmission; however, expansion occurs far more commonly in paternal transmission and contraction occurs more commonly in maternal transmission [Wheeler et al 2007].

Offspring who inherit a:

- **Reduced-penetrance allele** (36-39 CAG repeats) are at risk for HD but may not develop symptoms. Asymptomatic individuals with CAG repeats in this range are common [Kay et al 2016, Ibañez et al 2024]. The loss of CAA and CCA trinucleotide interruptions has been shown to modify the probability of disease in individuals with 36-39 CAG repeats [Wright et al 2019, Findlay Black et al 2020, Dawson et al 2024] (see Penetrance).
- **Full-penetrance allele** (40 or more CAG repeats) are at risk of developing HD with increased certainty assuming a normal life span (see Genotype-Phenotype Correlations and Penetrance).
- Each child of an affected individual who is homozygous for CAG repeat expansion in *HTT* will inherit an HD-causing allele.

Other family members. The risk to other family members depends on the genetic status of the proband's parents: if a parent is affected and/or has an expanded CAG repeat in *HTT*, the parent's family members are at risk.

Related Genetic Counseling Issues

Predictive testing (i.e., testing of asymptomatic at-risk individuals). Testing of asymptomatic adults at risk for HD is possible but requires careful deliberation, as there is currently no cure for the disorder [Almqvist et al 2003, Harper et al 2004, Hawkins et al 2011, Squitieri et al 2003]. Testing for a CAG trinucleotide repeat expansion in *HTT* in the absence of definite symptoms of HD is predictive testing. Correlations between the number of CAG repeats and the age of onset of HD (see supplementary tables at [ubc.ca](https://www.ubc.ca) [pdf]) and some clinical aspects of the disorder (see Genotype-Phenotype Correlations) have been described. When testing at-risk individuals for HD, it is helpful to test for the *HTT* CAG repeat expansion in an affected family member to confirm that the disorder in the family is HD.

- At-risk asymptomatic adult family members may seek testing in order to make personal decisions regarding reproduction, financial matters, and career planning. Asymptomatic individuals at risk may also be eligible to participate in clinical trials. Others may have different motivations including simply the "need to know."
- Testing of asymptomatic at-risk adult family members usually involves pretest interviews in which the motives for requesting the test, the individual's knowledge of HD, the possible impact of positive and negative test results, and neurologic status are assessed. Those seeking testing should be counseled about problems they may encounter with regard to health, life, and disability insurance coverage, employment and educational discrimination, and changes in social and family interaction. Interestingly, a study has found that genetic testing does not increase the risk for discrimination; perceived genetic discrimination is more likely due to the family history of HD regardless of genetic status, rather than due to the specific results of the HD genetic test [Bombard et al 2009]. Other issues to consider include implications for the at-risk status of other family members [Bombard et al 2012]. Depression and suicidal ideation are issues to be addressed as part of the predictive testing program for HD [Robins Wahlin et al 2000, Robins Wahlin 2007]. Predictive testing should not be conducted if psychiatric issues are not managed prior to testing and if the health professional believes testing would be harmful (see [Genetic Testing Protocol for Huntington's Disease](#), a Huntington's Disease Society of America resource). Informed consent should be obtained and records kept confidential. Individuals with a pathogenic HD-causing allele need arrangements for long-term follow up and evaluations.

- Short-term follow up of the participants in the Canadian Predictive Testing Program has revealed that predictive testing for HD may maintain or even improve the psychological well-being of at-risk individuals even though some had negative experiences. About 10% of the group who were determined to be at decreased risk had serious difficulties adapting to their new status [Bloch et al 1992, Hayden 2003, Hawkins et al 2013]. The major issue for these individuals was the realization that they were facing an unplanned future. Overall, the demand for testing of at-risk asymptomatic adults has been lower than expected in studies conducted before the availability of direct molecular genetic testing. Consistent with use of medical services and genetic testing in general, women are more likely than men to undergo predictive testing for HD [Taylor 2005, Baig et al 2016].

The Huntington's Disease Society of America [Genetic Testing Protocol for Huntington's Disease](#) provides an in-depth review of genetic counseling considerations in predictive testing for HD.

Predictive testing in minors (i.e., testing of asymptomatic at-risk individuals younger than age 18 years) for typically adult-onset conditions for which early treatment would have no beneficial effect on disease morbidity and mortality should be discussed in the context of formal genetic counseling. The autonomy of the minor is a primary concern and consideration should be given to delay of predictive genetic testing until the at-risk individual is capable of informed decision making. Testing must be preceded by a complete neurologic and neuropsychological evaluation.

In a family with an established diagnosis of HD, it is appropriate to consider testing of symptomatic individuals regardless of age.

Considerations for individuals with genetic modifiers (e.g., loss of interruption [LOI] variants). Current diagnostic methods underestimate *HTT* CAG length by two CAG repeats in individuals with loss of the CAA interruption in the CAG repeat (CAG-CCG LOI variant and CAG LOI variant; see Figure 5) [Wright et al 2019, Findlay Black et al 2024]. This is particularly problematic for individuals with a CAG repeat length near the cutoff for diagnostic ranges such as the upper-intermediate range (34-35 CAG repeats) and reduced-penetrance range (38-39 CAG repeats), as the underestimation results in being classified into an inaccurate diagnostic category. This has major implications affecting genetic counseling, management, and family planning [Dawson et al 2024, Findlay Black et al 2024]. These findings introduce new challenges for genetic diagnosis, making it essential that counseling support be continually updated to reflect emerging evidence [Brunham & Hayden 2012, Semaka et al 2013, Semaka & Hayden 2014].

Considerations in families with an apparent *de novo* HD-causing allele. When neither parent has an HD-causing *HTT* allele (>35 CAG repeats) or an intermediate allele (27-35 repeats), nonmedical explanations including alternate paternity or maternity (e.g., with assisted reproduction) and undisclosed adoption could be explored.

Family planning

- The optimal time for determination of genetic risk and discussion of the availability of prenatal/preimplantation genetic testing is before pregnancy.
- Similarly, decisions about testing to determine the genetic status of at-risk asymptomatic family members are best made before pregnancy.

Prenatal Testing and Preimplantation Genetic Testing

Once an HD-causing CAG repeat in *HTT* has been identified in an affected family member, prenatal and preimplantation genetic testing are possible.

Preimplantation genetic testing exclusion protocols allow for testing of the embryo for couples in an at-risk family who do not wish to undergo predictive testing for the HD-causing allele themselves [Kaplun 2025, Raine et al 2025].

Differences in perspective may exist among medical professionals and within families regarding the use of prenatal and preimplantation genetic testing. While most health care professionals would consider decisions regarding prenatal and preimplantation genetic testing to be the choice of the parents, discussion of these issues is appropriate.

Resources

The following disease-specific and/or umbrella support organizations and/or registries are listed for the benefit of individuals with this disorder and their families. GeneReviews is not responsible for the information provided by other organizations. For information on selection criteria, click [here](#).

- **European Huntington's Disease Network (EHDN)**
Germany
ehdn.org
- **HDBuzz**
HDBuzz
- **Huntington Disease Africa**
Phone: 254746734559
Email: info@hd-africa.org
hd-africa.org
- **Huntington Society of Canada**
Canada
Phone: 800-998-7398
Email: info@huntingtonsociety.ca
huntingtonsociety.ca
- **Huntington's Disease Society of America**
Phone: 800-345-4372
Email: hdsainfo@hdsa.org
hdsa.org
- **La Société Huntington du Québec (Huntington Society of Quebec)**
Canada
Phone: 514-282-4272; 877-282-2444
Email: shq@huntingtonqc.org
huntingtonqc.org
- **Testing for Huntington Disease: Making an Informed Choice**
Booklet providing information about Huntington disease and genetic testing
University of Washington Medical Center
[Testing for Huntington Disease: Making an Informed Choice](#)
- **Hereditary Disease Foundation**
Phone: 212-928-2121
Email: cures@hdfoundation.org
hdfoundation.org

- **Enroll-HD Registry**
[Clinical Research](#)
- **Huntington Study Group (HSG)**
Phone: 800-487-7671
Email: info@hsglimited.org
huntingtonstudygroup.org

Molecular Genetics

Information in the Molecular Genetics and OMIM tables may differ from that elsewhere in the GeneReview: tables may contain more recent information. —ED.

Table A. Huntington Disease: Genes and Databases

Gene	Chromosome Locus	Protein	Locus-Specific Databases	HGMD	ClinVar
HTT	4p16.3	Huntingtin	HTT database	HTT	HTT

Data are compiled from the following standard references: gene from [HGNC](#); chromosome locus from [OMIM](#); protein from [UniProt](#). For a description of databases (Locus Specific, HGMD, ClinVar) to which links are provided, click [here](#).

Table B. OMIM Entries for Huntington Disease ([View All in OMIM](#))

143100	HUNTINGTON DISEASE; HD
613004	HUNTINGTIN; HTT

Molecular Pathogenesis

HTT encodes the huntingtin (HTT) protein, which contains a polyglutamine (polyQ) stretch at its amino terminus [Saudou & Humbert 2016]. HTT is required for early embryonic development and plays a crucial role in neurogenesis [Barnat et al 2020]. HTT acts as a scaffold protein that facilitates several essential functions in the cell, including cytoskeletal dynamics, autophagy, endocytosis, vesicular trafficking (brain-derived neurotrophic factor), synaptic transmission, energy metabolism, calcium transport, RNA processing transcription/translation, chromatin remodeling, and DNA repair [Jimenez-Sanchez et al 2017, Kennedy et al 2022, Alteen et al 2023, Justice et al 2025].

The pathogenesis of Huntington disease (HD) remains poorly understood, partly because HTT is ubiquitously expressed and its diverse functions are not fully understood. In the presence of a CAG expansion, many of these cellular roles are disrupted, but because of the number of interactions it makes it difficult to determine what is contributing to disease pathogenesis (see Figure 7) [Saudou & Humbert 2016].

The CAG trinucleotide repeat expansion in exon 1 of *HTT* is translated into an uninterrupted stretch of polyQ residues, resulting in the production of mutant HTT (mHTT) protein. The expanded polyQ tract confers a toxic gain of function, resulting in the dysregulation of vital cellular processes, which ultimately leads to protein aggregation and cell death [Hana et al 2024]. Large intracellular aggregates composed of mHTT, known as inclusion bodies, have been found throughout postmortem brains and are considered a pathologic hallmark of HD [DiFiglia et al 1997]. N-terminal fragments containing exon 1 of *HTT* with expanded polyQ repeats, generated by either aberrant splicing or proteolytic cleavage, have been observed in human postmortem brains and are associated with higher toxicity compared to the full-length protein [DiFiglia et al 1997, Sathasivam et al 2013, Neueder et al 2017].

Mounting evidence supports the idea that somatic expansion of the expanded *HTT* CAG repeat plays a central role in HD pathogenesis. Somatic expansion within the striatum correlates strongly with earlier disease onset [Telenius et al 1995, Kennedy et al 2003, Shelbourne et al 2007, Swami et al 2009, Mouro Pinto et al 2020]. Genetic studies have revealed that variants in DNA maintenance genes modify HD onset, supporting the concept that repeat length-dependent instability contributes to neuronal vulnerability. It has been proposed that longer CAG repeats exhibit an increasing propensity to expand further, eventually surpassing a toxic threshold that triggers neuronal dysfunction [Kaplan et al 2007]. In this model, somatic instability acts as a catalyst for expansion, while a subsequent toxic gain-of-function mechanism drives cell death once the repeat length crosses a critical threshold.

Additional mechanisms at an RNA level have been proposed, including missplicing of exon 1 fragments, RNA foci formation, repeat-associated non-AUG (RAN) translation, and ribosome stalling resulting in repression of protein synthesis, although their contribution to adult-onset HD remains uncertain [Wang et al 2011, Bañez-Coronel et al 2015, Neueder et al 2017, Eshraghi et al 2021].

Table 9. *HTT* Technical Considerations

Technical Issue	Comment [Reference]
Sequence of repeat	CAG; however, an interrupted repeat, [(CAG) <i>n</i> -CAA-CAG], is present on the majority of alleles [Wright et al 2019]. For clinical implications see Genotype-Phenotype Correlations.
Methods to detect expanded allele (see Table 10)	• Conventional PCR, triplet-primed PCR (TP-PCR) [Jama et al 2013], & Southern blotting [Bean & Bayrak-Toydemir 2014] have been described. • Current diagnostic assays are only capable of measuring CAG length up to the penultimate CAA in the canonical sequence. In the presence of the loss of the CAA interruption (CAA to CAG), current diagnostic methods underestimate the CAG repeat length by 2 CAG repeats. While this does not affect the majority of individuals w/HD in the full-penetrance range, it is particularly relevant for those in the reduced-penetrance range, where the LOI is enriched [Wright et al 2019, Findlay Black et al 2020, Dawson et al 2024].
Somatic instability	Alleles w/abnormal number of CAG repeats may display somatic instability of the repeat [Telenius et al 1994].
Germline instability	Although expansion & contraction of repeat length can occur w/maternal or paternal transmission, expansion occurs far more commonly in paternal transmission, w/large expansions (i.e., >7 CAG repeats) occurring almost exclusively through paternal transmission [Semaka et al 2013a]. Most persons w/juvenile HD inherit the expanded allele from their father [Nahhas et al 2005].

HD = Huntington disease; LOI = loss of interruption

Methods to characterize *HTT* CAG repeats. Due to the technical challenges of detecting and sizing *HTT* CAG repeat expansions, multiple methods may be needed to rule out or detect CAG repeat expansion (see Table 10). Most repeats may be detected by traditional PCR. However, detection of apparent homozygosity for a normal CAG repeat does not rule out the presence of a very large expanded CAG repeat. Thus, testing by triplet-primed PCR (TP-PCR) or Southern blotting may be required. In addition, somatic and germline instability of expanded repeats must be considered.

Table 10. Methods to Characterize *HTT* CAG Repeats

Interpretation of CAG Repeat Number	Expected Results by Method		
	Conventional PCR	Triplet-primed PCR ¹	Expanded repeat Analysis ²
Normal: ≤26	Detected ³	See footnote 1.	Expansions can be detected & repeat size can be approximated. ⁶
Intermediate: 27-35	Detected ^{3, 4}		
Pathogenic (reduced penetrance): 36-39	Detected ^{3, 4}		
Pathogenic (full penetrance): ≥40	Most alleles detected ^{3, 4}	Expansions detected but repeat length may not be determined ⁵	

1. The design of a triplet-primed PCR (TP-PCR) assay may include conventional PCR primers to size normal repeats and detect expanded repeats in a single assay. The TP-PCR assay itself does not determine repeat size, even alleles in the normal range.
2. Methods to detect and approximate the size of expanded repeats include long-range PCR sized by gel electrophoresis and Southern blotting. The upper limit of repeat size detected will vary by assay design, laboratory, and sample due to competition by the normal allele during amplification.
3. Detection of an apparently homozygous repeat does not rule out the presence of an expanded CAG repeat; thus, testing by TP-PCR or Southern blot analysis is required to detect a repeat expansion.
4. PCR-based methods detect alleles up to about 115 CAG repeats [Potter et al 2004, Levin et al 2006]. Other methods may occasionally be useful to identify large CAG repeat tracts. The upper limit of repeat size detected will vary by assay design, laboratory, sample, and/or patient due to competition by the normal allele during amplification.
5. TP-PCR may indicate the presence of a highly expanded allele without precise estimation of the CAG repeat length.
6. Southern blotting for the CAG repeat expansion has been described [Bean & Bayrak-Toydemir 2014].

Mechanism of disease causation. Available evidence supports a predominantly gain of function mechanism, given that expression of an expanded polyglutamine is toxic itself, but the exact mechanism of toxicity is still uncertain. The contribution of the loss of function of *HTT* protein cannot be discounted because deletion or inactivation of *HTT* can lead to neurodegeneration.

Table 11. *HTT* Pathogenic Variants Referenced in This *GeneReview*

Reference Sequences	DNA Nucleotide Change	Predicted Protein Change	Repeat Range
NM_002111.8 NP_002102.4	c.52CAG[9_26]	p.Gln18[9_26]	Normal
	c.52CAG[27_35]	p.Gln18[27_35]	Intermediate
	c.52CAG[36_39]	p.Gln18[36_39]	Reduced penetrance
	c.52CAG[40_?]	p.Gln18[40_?]	Full penetrance

Variants listed in the table have been provided by the authors. *GeneReviews* staff have not independently verified the classification of variants.

GeneReviews follows the standard naming conventions of the Human Genome Variation Society (varnomen.hgvs.org). See [Quick Reference](#) for an explanation of nomenclature.

Chapter Notes

Author Notes

Booklets available through the [Huntington Society of Canada](#):

- Loss and Grief: Coping with the Death of a Loved One and with Other Losses Related to Huntington Disease
- A Physician's Guide to the Management of Huntington Disease (3rd edition)

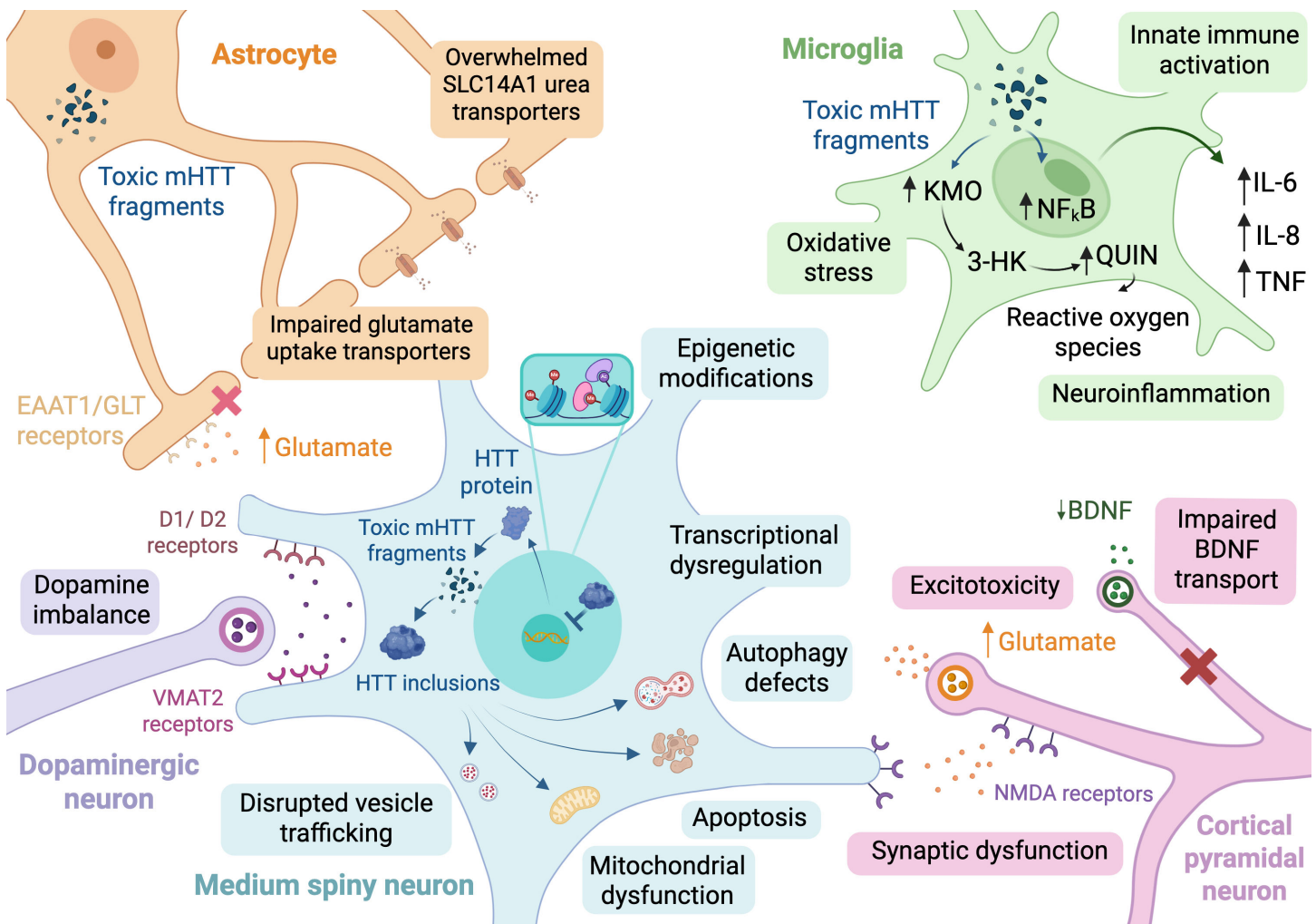


Figure 7. Potential pathogenetic cellular mechanisms in Huntington disease (HD)

Schematic overview of the mechanisms occurring in astrocytes (orange), microglia (green), dopaminergic neurons (purple), medium spiny neurons (blue), and cortical pyramidal neurons (pink).

3-HK = 3-hydroxykynurenine metabolite; BDNF = brain-derived neurotrophic factor; D1/D2 receptors = dopamine receptors D1 and D2; EAAT1/GLT = excitatory amino acid transporter 1 / glutamate transporter; KMO = kynurenine 3-monooxygenase enzyme; mHTT = mutant huntingtin; NF κ B = nuclear factor kappa-light-chain-enhancer of activated B; NMDA = N-methyl-D-aspartate receptor; QUIN = quinolinic acid; SLC14A1 = solute carrier family 14 member 1; VMAT = vesicular monoamine transporter

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- Caregiver's Handbook for Advanced Stages of Huntington Disease
- Juvenile Huntington Disease: A Resource for Families, Health Professionals and Caregivers
- Understanding Behaviour in Huntington Disease: A Guide for Professionals (3rd edition)
- Personal Perspectives on Genetic Testing for Huntington Disease
- Understanding Huntington Disease: A Resource for Families

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Revision History

- 12 February 2026 (sw) Comprehensive update posted live
- 11 June 2020 (sw) Comprehensive update posted live
- 5 July 2018 (sw) Comprehensive update posted live
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- 22 April 2010 (me) Comprehensive update posted live
- 19 July 2007 (me) Comprehensive update posted live
- 15 February 2005 (me) Comprehensive update posted live
- 23 October 1998 (pb) Review posted live
- 16 May 1998 (mh) Original submission

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